Regeneration and fuel loading with varying overstory retention in redwood stands 10 years after transformation to multiaged management

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Judson Fisher

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Committee Membership

Dr. Pascal Berrill, Committee Chair

Dr. Jeffrey Kane, Committee Member

Dr.Rosanna Overholser, Committee Member

Dr. Andrew Kinzinger, Program Graduate Coordinator

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ABSTRACT

Regeneration and fuel loading with varying overstory retention in redwood stands 10 years after transformation to multiaged management

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Forest managers using retention forestry in western forests need information on regeneration dynamics and fuel development to design stands that are both productive and fire resilient. This study analyzes 10-year remeasurement data from the Redwood Multi-aged Experiment in Jackson Demonstration State Forest, California, comparing regeneration and fuel responses across group selection (GS), aggregated (HA), and dispersed (LD, HD) retention treatments. Measurements included sprout growth and size of redwood and tanoak, seedling density, and surface fuel loading.

Redwood responded strongly to increasing canopy openness, with basal area and height greatest in GS and lowest in HD. Tanoak showed a more modest response, consistent with greater shade tolerance, and differences between species narrowed under higher retention. Redwood also grew faster than tanoak across all treatments, though growth declined over time for both species. Douglas-fir regeneration was low and did not vary among treatments.

Fuel loads showed few differences among treatments prior to thinning, but vegetative fuels were highest in more open treatments. Pre-commercial thinning reduced live fuels and increased fine dead fuels, particularly in treatments that had accumulated greater biomass.

These results highlight a key trade-off: treatments that promote rapid redwood growth also increase short-term surface fuel loads following thinning. Effective multiaged management in redwood systems therefore requires integrating regeneration objectives with fuel treatment strategies over time.

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I acknowledge that this research was conducted on the unceded territory of the Northern Pomo tribes, the original stewards of these lands. I recognize and deeply regret my failure to engage with the tribe during my fieldwork. This omission carries significant implications for the responsible management of these forest ecosystems, particularly regarding the health and development of tanoak, a keystone species and a principal food source sustaining the Pomo people since time immemorial. I understand the importance of incorporating Indigenous knowledge into ecological research and management, and I sincerely hope that future projects on these lands will be conducted in genuine and collaborative partnership with Pomo tribal members. I believe their

expertise is essential to ensuring the respectful and sustainable stewardship of their ancestral territory.

I would like to thank my children: Fox and Hazel, and my partner Megan for their support and patience in allowing me the time and space to continue working on this thesis even when it went on much longer than I had originally anticipated and meant the sacrifice of some of our precious shared weekend time. I'm deeply grateful for their willingness to prioritize my personal development and their confidence in my eventual success.

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1. INTRODUCTION

1.1 Multiaged management

Ecological forestry, which maintains a wide range of ecosystem services while also supplying timber requires a diverse landscape of highly varied forest structures (Aplet, 1994; Nolet et al., 2018; O'Hara, 2001). These, in turn, require a variety of silvicultural techniques to implement and sustain (O'Hara, 1998; Schütz, 2002). The development of multiaged stand structures has long been of interest to silviculturists as a key alternative to the conceptually and logistically simpler, even-aged management (Schütz, 1999). Multiaged silviculture refers to the retention of trees of distinctively different age classes, growing together within the same stand. These cohorts may co-occur at the tree level, or in small, even-aged patches within the stand. In the latter case, the distinction between even- and multiaged management can become blurred with increasing patch size, but patches are generally much smaller (often less than 1 ha) than the stands they compose. The pursuit of multiaged stand structures has often met with mixed results (O'Hara, 2002) and this has led to the investigation of several different systems for achieving such structures. Research into the efficacy and results associated with these is ongoing (Beese et al., 2019; Nolet et al., 2018). One such system that has gained popularity in recent decades is known as the retention system, which allows for the retention of a range of tree densities, in dispersed or aggregated spatial patterns and can be used to maintain multiaged stands or transform from even-aged management (S. J. Mitchell & Beese, 2002).

Redwood forests offer a prime opportunity for multiaged management because coast redwood (*Sequoia sempervirens*) regenerate reliably via stump sprouting and are relatively shade tolerant. The very high leaf areas observed in these forests, suggest their suitability for a multi-layered forest structure (Berrill & O'Hara, 2007; Van Pelt et al., 2016). Additionally, with their high timber value and productivity, redwood forests are of keen interest to private timber producers. Despite redwood's fitness for multiaged stand structures, the successful development of subordinate cohorts depends on adequate access to light, and light deficiency can lead to reduced vigor and mortality in young sprouts and understory trees (Barrett, 1988; Muma et al., 2022; O'Hara et al., 2007; Webb et al., 2012).

Complicating redwood regeneration and sprout development is the fact that the competing hardwood species, tanoak (*Notholithocarpus densiflorus*), is also shade tolerant and a vigorous resprouter. Tanoak is a keystone species in terms of wildlife habitat and First Nation's cultural identities, but from a timber production standpoint it is often perceived as a nuisance due to a lack of market development combined with its widespread proliferation following intensive, repeated conifer harvesting (Bowcutt, 2011). While redwood grows more quickly than tanoak in multiaged stands, competition from hardwoods such as tanoak reduces conifer growth and drought resistance (Berrill et al., 2021; Dagley et al., 2023). Meanwhile, there has been little scientific investigation into the development of tanaok under shade (Waring & O'Hara, 2008; Wilkinson et al., 1997).

1.2 Management effects on sprouting

1.2.1 Measures and general patterns of sprout development over time

Tree sprouting response following cutting is commonly characterized by presence (percent stumps sprouting), abundance (sprout density), and size (height and diameter). Interpretation of these metrics depends on the time between treatment and measurement, as early sprout growth is largely driven by mobilization of carbohydrates stored in the parent stem and root system (Del Tredici, 2001). As a result, differences associated with external conditions may not be evident early in development. Redwood sprout clumps may consist of more than 100 stems in the first year after cutting (Neal, 1967), but rapidly self-thin, particularly under high light conditions (Boe, 1975). Under overstory competition, self-thinning may proceed more rapidly and can result in mortality of entire sprout clumps (O'Hara & Berrill, 2010). In contrast to eastern hardwoods, where self-thinning typically occurs over decades (Gould et al., 2007), this process may extend for centuries in long-lived redwoods (O'Hara et al., 2017).

Because percent sprouting, sprout density, and sprout size represent distinct aspects of sprouting response, they vary with species, site characteristics, overstory density, parent stump size or age, and geographic context. Even when these factors are considered, substantial unexplained variation among sites may remain (Keyser & Loftis, 2015; Nieves et al., 2022).

1.2.2 Factors influencing initial sprout response

Across many sprouting species, the proportion of stumps that sprout following cutting often declines with increasing tree size or age, although this pattern varies by species

and site conditions (Johnson, 1977; Nieves et al., 2022). In redwoods, evidence for declining percent sprouting with increasing size or age is mixed (Barrette, 1966; Lindquist, 1979; Neal, 1967; Wiant & Powers, 1967), potentially reflecting the wide range of tree sizes and ages present in these forests. Some studies suggest that percent sprouting may increase with age to a threshold and decline in trees older than approximately 200–400 years (O'Hara et al., 2007; Powers & Wiant, 1970).

After cutting, 90–100% of second-growth redwoods (< 90 cm DBH) typically sprout (Barrette, 1966; Lindquist, 1979), whereas sprouting among larger, older redwoods may approach 50% (Boe, 1975; Neal, 1967). Declining sprouting with increasing size has also been documented in tanoak and other coastal hardwoods (Harrington et al., 1992).

Residual overstory density may influence percent sprouting in some species and regions, but evidence for this effect is inconsistent and sensitive to study design (Nieves et al., 2022). In redwood forests, this effect is generally weak or absent (Barrett, 1988; Lindquist, 1979). The number of sprouts produced per stump is also typically unrelated to overstory density (Atwood et al., 2009; Knapp et al., 2017) and is assumed to follow a similar pattern in redwood forests (Lindquist, 1979; O'Hara & Berrill, 2010).

1.2.3 Internal and external factors influencing sprout development over time

Over time, sprout growth and survival are strongly influenced by competitive conditions. Among sprouting species, sprout growth generally increases with understory light availability (Gardiner & Helmig, 1997; Keyser & Zarnoch, 2014; Knapp et al., 2017), although the influence of light is weakest early in development, when growth is dominated by stored carbohydrates (Gardiner & Helmig, 1997; Keyser & Loftis, 2015).

Despite its shade tolerance, redwood requires a minimum light threshold to sustain sprout growth (O'Hara & Berrill, 2010), and growth rates are sensitive to understory light (Berrill et al., 2018).

Sprout growth is also positively related to stump diameter, with larger stumps producing faster-growing sprouts, a pattern observed in redwood (Berrill et al., 2018), tanoak (Harrington et al., 1992), and eastern hardwoods (Dey et al., 1996; Keyser & Loftis, 2015). Survival of sprouts in subsequent years depends on overstory density and competitive pressure, particularly as stands approach canopy closure. Entire sprout clumps may experience rapid mortality under low-light conditions (O'Hara & Berrill, 2010). Longer-term sprout persistence has also been shown to vary with site and regional factors (Keyser & Loftis, 2015; Nieves et al., 2022), although these influences have not been well quantified in redwood forests.

Due to rapid early growth, sprouting species can influence the composition of regenerating stands (Del Tredici, 2001), sometimes favoring less desirable species (Keyser & Zarnoch, 2014). In redwood ecosystems, redwood sprouts typically outgrow tanoak stump sprouts during the first five years following partial harvest (Muma et al., 2022), though the longer-term implications of these dynamics for stand development remain uncertain (Berrill et al., 2018). Additional disturbance interactions, including herbivory and wildlife damage, may further influence sprout development and species composition (Berrill et al., 2017; Dagley et al., 2018; Schneider et al., 2023; Wilkinson et al., 1997).

1.3 Forest fuels

Throughout many of the fire-adapted forests of California, fire exclusion combined with timber harvest has led to dense, suppressed stands, proliferation of more fire-sensitive species, and an accumulation of surface fuels (Safford & Stevens, 2017; Stephens et al., 2009). This situation combined with climate change has led to increased size and frequency of high-severity fires in many regions (Parks & Abatzoglou, 2020; Westerling, 2016), prompting widespread interest in fuel treatments and silvicultural interventions creating resilient stand structures.

This interest has seen less momentum in the redwood region, likely due to the perceived safety of these typically moist forests from the threat of large wildfires. Yet redwood litter is among the most flammable of conifer litter types (Fonda et al., 1998), seasonal drought leads to cured fuels, especially during extended breaks in coastal fog (Jacobs et al., 1985), and numerous physiological adaptations suggest that redwood has evolved under fire disturbance pressure (Varner & Jules, 2017). More concretely, there have been at least six large fires in redwood ecosystems since 2003, burning at least 189,000 ha including widespread areas of canopy loss. Scientific consensus places the pre-colonization fire return interval for redwood forests at 6-25 years across their range (Lorimer et al., 2009). It is assumed that much of this activity is attributable to indigenous burning (Varner & Jules, 2017).

Consistently low foliar moisture content levels have been recorded for tanaok (Kuljian & Varner, 2010) and the foliage has been compared to that of other sclerophyllous species such as those found in chaparral ecosystems which are

characterized by intense, stand replacing fires suggesting high flammability, particularly with low canopy base heights (Fryer, 2008; McDonald, 1981). And tanoak leaf litter is among the most flammable of western hardwoods and has been compared to that of fire adapted conifers (Varner et al., 2017).

There have been several studies that have quantified various fuel strata in redwood forests. Kittredge (1940) did so for duff and litter in a redwood plantation. Greenlee (1983) studied fuels at Big Basin State Park. Stuart (1985) reported on fuels at Humboldt Redwoods State Park. Finney and Martin (1993a, 1993b) reported on fuels in second-growth redwood forests (aged ~100 years) at Annadel and Humboldt Redwoods State Parks. Graham (2009) reported on fuels in old growth stands across redwood's range. Glebocki (2015) studied fuels with and without thinning treatments in young (< 50 years) redwood/Douglas-fir stands. No fuel studies, to my knowledge, have been conducted in redwood forests actively managed with multiaged silviculture, but fuel dynamics represent a potentially important decision variable to consider when managing forest stands that may be subjected to intentional or unintentional fire.

1.3.1 Management effects on fuels

Depending on the method used, thinning and harvest treatments may increase, or not affect surface fuel loading. Whole tree removal results in the least fuel accumulation but can be more expensive than other options (Han & Han, 2020). Most other treatment methods increase surface fuels (Agee & Skinner, 2005; Stephens et al., 2009). The magnitude of this increase is variable, reflecting factors such as treatment mode, intensity, and pre-existing conditions (Schwilk et al., 2009). Additional research is needed to clarify the effects of these factors on short-term (Hood et al., 2020; Schwilk et

al., 2009; Stephens et al., 2009) and long-term changes to surface fuel load resulting from specific management actions (Hood et al., 2020; Stephens et al., 2012).

The majority of fine dead fuels (< 8 cm) generated by treatment activities typically decompose within 10 years (Burton et al., 2022; Hood et al., 2020; Martinson & Omi, 2013; O'Hara et al., 2017; Stephens et al., 2012). But live woody fuels, which respond vigorously to increased growing space, often persist or increase over time (Keyes & Varner, 2006). The nature of this response depends on eco-type and the amount of growing space liberated by the treatment, which in turn can become occupied by herbaceous plants (Vilà-Vilardell et al., 2023), shrubs (Odland et al., 2021), or small trees (Hood et al., 2020).

Overtime, duff and litter loads are frequently lower in more open stands than stands with a more closed canopy. This may result from increased decomposition rates due to greater insolation and increased throughfall, or reduced deposition rates resulting from fewer canopy fuels (Hood et al., 2020; Keane, 2008).

Most fuel reduction thinning research focuses on ponderosa pine (*Pinus ponderosa*) forests in the United States, with additional studies from other Mediterranean and semi-arid regions (Burton et al., 2022; Schwilk et al., 2009; Vilà-Vilardell et al., 2023). Far fewer studies have been conducted in coastal forests (Norman et al., 2009), but see Wilder et al. (2025).

1.4 Objectives

This research evaluates regeneration and fuel dynamics in managed coast redwood forests located within the Jackson Demonstration State Forest in Mendocino County,

California. The study utilizes a 10-year re-measurement of a replicated multiaged silviculture experiment to compare the effects of four distinct harvesting scenarios: group selection (GS), high-density aggregated retention (HA), and high- and low-density dispersed retention (HD; LD). The specific objectives of this study were to:

1. Quantify the development of live understory and dead surface fuels across the range of residual overstory densities provided by these treatments.
2. Evaluate the growth and survival of redwood and tanoak stump sprouts to determine how overstory retention influences the development of the next cohort.
3. Assess the impact of pre-commercial thinning (PCT) on surface fuel profiles.

By analyzing these interactions, this research explores the fundamental trade-offs and synergies between promoting cohort productivity and managing fuel hazards, providing information necessary to design forest structures that are both productive and fire-resilient (North et al., 2021; York et al., 2021), such as has been practiced by indigenous people in this region for millennia (Anderson, 2013), and, in the American southeast for around 100 years (R. J. Mitchell et al., 2009).

2. METHODS

2.1 Site description

Situated within the unceded ancestral territory of the Northern Pomo, near the coast of northern California, the 20,000-ha Jackson Demonstration State Forest (JDSF) was established in 1947 after being cut over multiple times starting in 1862. The mission of the JDSF is to remain in timber production for research and demonstration purposes. The predominant cover type is redwood, tanoak, and Douglas-fir forest at elevations ranging from 20 m near the coast to 700 m inland. The climate is Mediterranean with cool, wet winters and an average annual precipitation of about 1150 mm. The sites that compose my experiment were all redwood dominated with significant components of tanoak and Douglas-fir and were well stocked with trees 80 to 100 years old. All my study sites are between 13 and 16 km from the Pacific Ocean at elevations between 130 and 300 m, on well-drained, sandstone-derived soils. The sites are located on various aspects on mid to upper-slope positions (Figure 1), and the site quality is classified as Site Class II (Lindquist & Palley, 1963; Webb et al., 2012).

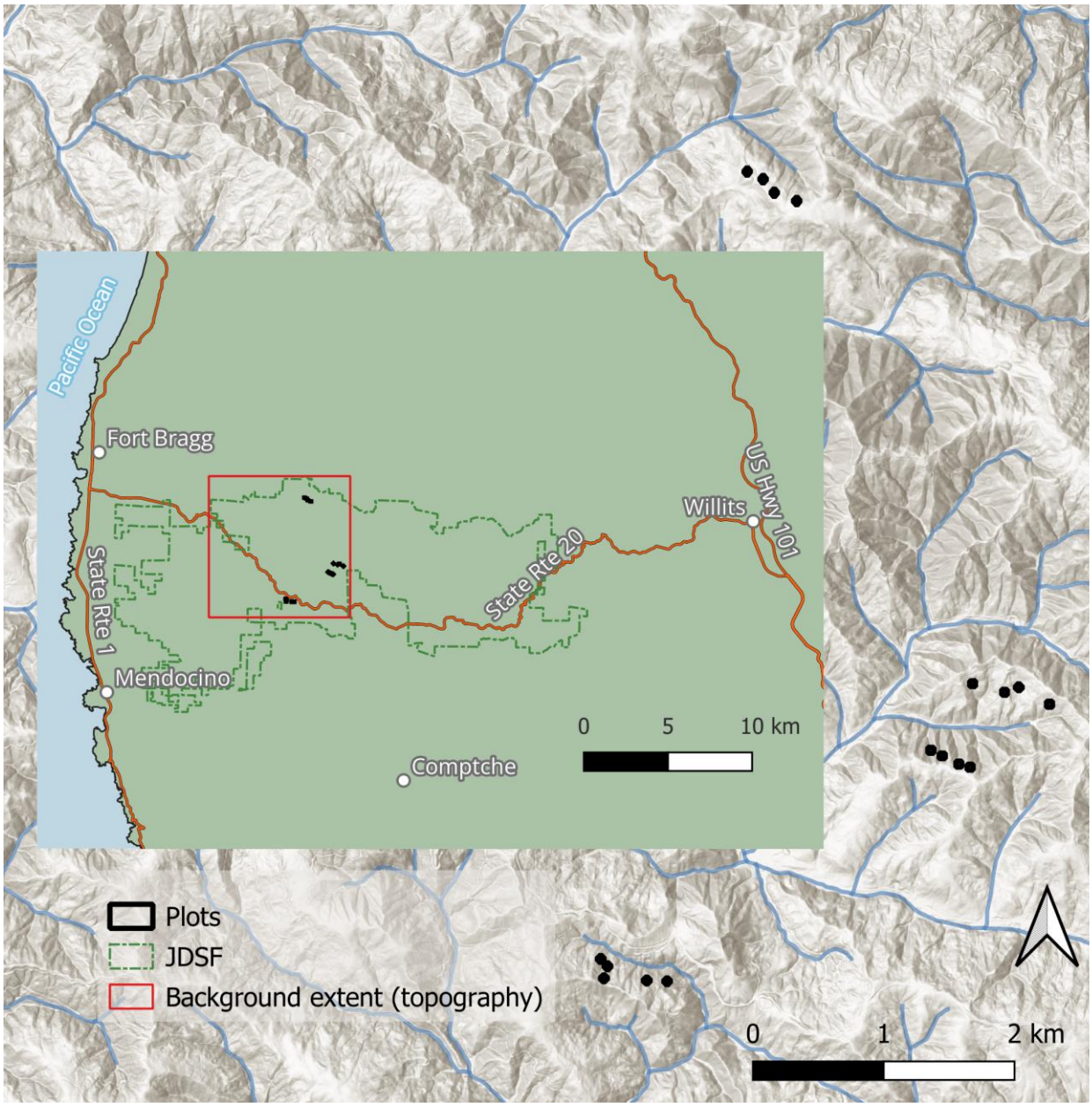


Figure 1. The vicinity, plot locations, and topography for my study plots. The small red box in the inset delineates the area of the topographical map seen in the background. Study plots appear in inset and background as black dos. The Jackson Demonstration State Forest (JDSF) comprises much of the Rte 20 corridor between Willits and the Pacific Ocean in Northern California. My study plots can be seen on mid- to upper-slope positions on a mixture of north- and south-facing aspects around the central portion of JDSF.

2.2 Study design

In 2012, four treatments were replicated at four different sites. The treatments were: group selection (GS), high-density dispersed retention (HD), high-density aggregated retention (HA), and low-density dispersed retention (LD). The GS treatment is composed of a circular 1-ha opening in which all trees were removed, surrounded by a 50-meter buffer of light thinning in which less than 1/3 of the basal area was removed. The remaining treatments were applied over 2-hectare treatment units.

The target post-harvest density was specified in terms of relative density, defined as the proportion of the assumed total site carrying capacity based on an upper stand density index (SDI) limit for redwood of 2,470 stems ha^{-1} (Reineke, 1933). The HA and HD treatments targeted a residual overstory relative density of 20%, which was selected as an upper limit to support the objective of maintaining conifer growth. In contrast, the LD treatment targeted a residual overstory relative density of 13% to favor establishment of a new cohort. These target densities were chosen to balance stand-level volume production with individual tree growth, assuming a subsequent, similar harvest entry after 20 years (Berrill & O'Hara, 2009).

After harvesting, a single 0.2-hectare square macro plot was established in the center of each treatment unit. Approximately 25 redwood and 25 tanoak sprout clumps that were well distributed across the plot, were selected for measurement. Additionally, for redwood, sprout clumps were selected evenly from those with, and without residual standing trees.

Ten years after the initial harvest, regeneration plots and fuel transects were established within the macro plot (Figure 2). Circular regeneration plots with a 4-meter radius were established 10 meters from each macro plot corner, towards the plot center.

Terminating at the center of each of these regeneration plots, 10-meter fuel transects were established parallel to the macro plot edges. Live fuels were estimated in 1-meter radius “sampling cylinders” at the five-, and nine-meter locations along each transect.

Following year 10 measurements, all plots received a PCT treatment where the objective was to release merchantable conifers such as redwood and Douglas-fir from competition. Competing trees and shrubs were cut, lopped, and scattered within the measurement plots and fuel transects were re-measured.

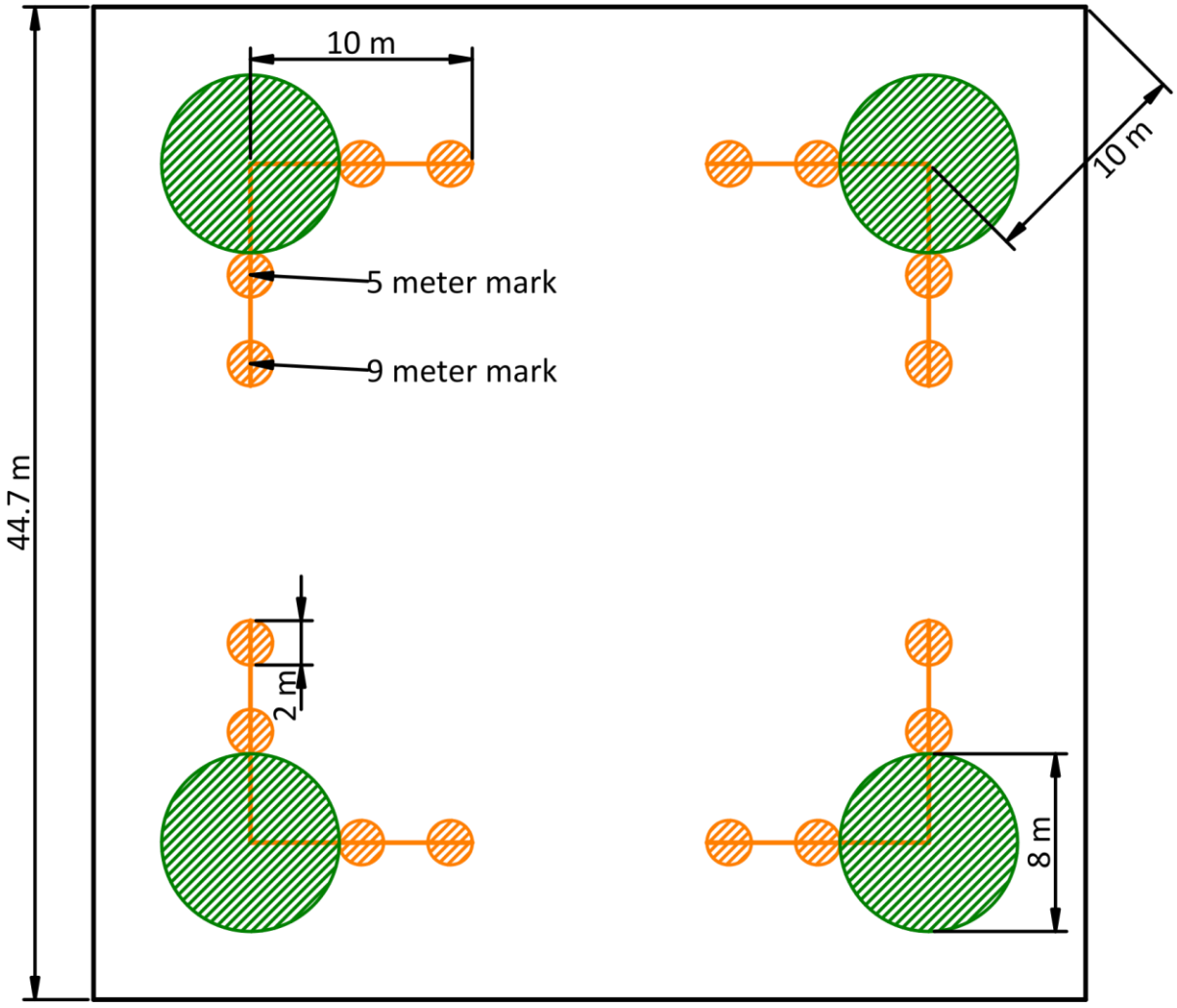


Figure 2. A diagram of my sampling design. The 0.2-ha macro plot is represented by a black square outline. Four 4-m radius regeneration plots are represented by green circles, inset from the corners of the macro plot. Ten-meter long transects are shown as orange lines emanating at right angles from the center of each regeneration plot. And 1-m radius sampling cylinders are shown as orange circles near the center and outside end of the transects.

2.3 Data collection

2.3.1 Regeneration

Ten years after the initial harvest, four regeneration plots (4-m radius) were established within each macro plot (Figure 2). All regenerating trees (sprouts and seedlings) within the plot were recorded. Regenerating trees less than 5 cm diameter at breast height (1.4 m) were tallied by 2.5 cm size classes and the actual DBH of all other regenerating trees were recorded. This allowed for an estimation of sprout and seedling density by species.

2.3.2 Sprout heights

Sprout height data were collected during the winter at years 1, 5, and 10 following the initial harvest. Sprout heights at each period were based on the tallest individual within the selected redwood and tanoak clumps at the time of measurement with a height pole placed at the same location on the ground at each measurement period. The observer eliminated parallax by observing from a location up-slope, level with the height of the sprout. Sprout DBH for only redwood was collected in year 10.

2.3.3 Fuels

Also at 10 years after the initial harvest, fuel transects were established and sampled according to the FIREMON protocol (Lutes et al., 2006), with some adaptations that were unique to this project. This protocol is very similar to that of Brown (1974). One-, 10-, 100-hr, and 1000-hr (or coarse woody) fuels were those less than 0.635 cm, 0.635 to 2.54 cm, 2.54 to 7.62 cm and those 7.62 cm and greater, respectively and included only dead and downed woody fuels (not those attached to live vegetation, or standing dead). For pre-PCT fuel measurements, transect lengths were two, two, and four meters for 1-, 10-, and 100-hr fuels, respectively. For post-PCT measurements, the length of the 1-hr fuel transect was reduced to one meter because of the dramatic

increase in these fuels. Coarse woody fuels were measured along the entire 10-meter transect.

Redwoods, like other species in Cupressaceae, shed branchlets instead of individual leaves. This leads to difficulty in distinguishing “litter” from 1-hr fuels. We chose to use an estimated cutoff of about 2 mm for redwood branchlets, where larger pieces were considered 1-hr fuel and smaller pieces were considered part of the litter layer. We did not distinguish between “leafy” and “awl like” leaf shapes (Graham, 2009).

Live vegetation percent cover and average height were estimated in 1-meter-radius “sampling cylinders” as described in the FIREMON protocol (Lutes et al., 2006). A notable exception is that the sampling cylinders were allowed to extend to the (average) top height of the live fuels that were continuous within less than a meter of the ground. This resulted in average live vegetation heights that could sometimes reach near the height of the sprouts. Average live fuel heights that were above two meters were estimated with the help of a clinometer. Live vegetation percent cover was estimated for four vegetation classes: live woody fuels, dead woody fuels, live herbaceous fuels, and dead herbaceous fuels. Dead “live” fuels included dead fuels attached to live plants, or those still rooted in the ground. Particles in these conditions are not counted as downed woody fuels when tallying fine and coarse woody fuels along fuel transects.

Average vegetation height was recorded for two classes: the average height of all woody fuels and the average height of all herbaceous fuels. These estimates pertain to the particles present in the sampling cylinder, and not the area-average, thus a cylinder with only one percent cover, could still have an average height of two meters,

i.e., the empty space within the cylinder does not affect the average height. The average height and percent-cover estimates are visual estimations and efforts were made to compare these frequently to ensure that estimates among observers were consistent. Percent cover estimates were discrete, rounded to the nearest class in the set: 0.5, 3, 10, and all subsequent 10 percent increments up to 100 percent.

The depth of duff and litter was measured at one representative location within each of the two, 1-meter-radius sampling cylinders along each transect. When duff and litter conditions varied greatly in a single sampling cylinder, a location was chosen to represent the average of conditions within the cylinder.

Fuel bed depth was estimated as the average height of the fine and coarse woody debris components within each sampling cylinder. Theoretically, this included litter but not duff. This estimate was not described in the FIREMON protocol, but it was designed to be made in a similar manner as described for vegetation heights.

2.4 Analysis

An analysis was conducted in three stages, one for each of the three response categories: fuels, sprout size, and regeneration density. All analyses were conducted using R and attempts were made to document all data, decisions, and techniques within a Quarto notebook, which is published at https://fisher-j.github.io/multi-age. All response variables were analyzed using multi-level models to account for the inherent nesting structure of the data, including multiple measurements of sprouts over time, where applicable. Grouping levels were included in the models if their variance estimate was determined to be significant. If including a grouping level resulted in a meaningful

change in estimates or confidence intervals, the grouping level was kept. Because the overall objective was to detect differences between treatments, treatment was used as the primary independent variable. Model development and selection were carried out using Akaike information criterion, and visualization of (probability integral) transformed residuals (Hartig, 2022). Models were built using the R package GLMMtmb, which provides a consistent framework for exploring different response distributions and link functions as well as the ability to model variance as a function of predictors (Brooks et al., 2017).

Models were interpreted using estimated marginal means calculated with the R packages marginaleffects and emmeans (Arel-Bundock et al., 2024; Lenth, 2021). Predictions incorporating the effects of all model components were made on the response scale for a grid of predictors consisting of all unique combinations of factors in the data that were used in at least one component of a given model (we didn't have numeric predictors) except in the case of tree heights where predictions were made for each observation in the data set. Standard errors for the predictions were calculated using the delta method. Predictions were averaged across non-focal predictors (especially "random" effects) to obtain the population estimates of marginal means. These means are expected to be applicable to other treatments when conducted across equal proportions of sites with conditions similar to those used in my study. Notably, my population estimates assume equal proportions of north and south facing aspects.

The formulas below use β to represent fixed effects and α to represent random effects. Subscripts indicate the variable associated with each effect. Superscripted Greek letters indicate the model component where fixed or random effects are present

in more than one component (e.g., μ for the conditional model, π for the hurdle model, and ϕ for the dispersion model). A “hurdle gamma” distribution is used for models of basal area and fuels and is defined as

$$f(y_i \mid \pi_i, \mu_i, \phi_i) = \begin{cases} 1 - \pi_i & \text{if } y_i = 0 \\ \pi_i \cdot \text{Gamma}(y_i \mid \mu_i, \phi_i) & \text{if } y_i > 0 \end{cases}$$

.

2.4.1 Regeneration

2.4.1.1 Basal area

Sprout count data from my 4-meter-radius regeneration plots were converted to basal area per acre per species. Tallies of species by size class for sprouts less than 5 cm DBH were converted to diameters using the class midpoint (1.27 or 3.81 cm) before calculating basal area per acre. Minor species (i.e., those other than redwood, tanoak, and Douglas-fir) were combined into an “other” category. Missing species on a plot were made explicit by assigning a value of 0 m^2 ha^{-1} for that species and plot.

Basal area was selected as the metric for quantifying redwood and tanoak abundance because prolific sprout regeneration in both species was undergoing self-thinning. In this context, basal area provided a more informative measure than stem density, as it better captured treatment responses by reflecting the relative contribution of established stems rather than the transient abundance of sprouts. Although numerous metrics have been used to asses shade tolerance (Forrester et al., 2014), basal area was selected because it integrates both growth and survival of sprouts. This approach assumes that growth and survival are positively correlated in forests that do not experience prolonged harsh winters (Lin et al., 2002).

Composition of regeneration in terms of basal area per acre represented by each species in a 4-meter radius vegetation plot 10 years after treatment was modeled as a gamma distribution with a log link with fixed effects for treatment and species, and random intercepts for site x species interaction. Dispersion was modeled separately as a function of species, using a log link and the rate of zeros was modeled using the logit link, for each species as well:

$$Y_i \sim \text{hurdle_gamma}(\pi_i, \mu_i, \phi_i)$$

$$\log(\mu_i) = \beta_0^{\mu} + \beta_{\text{trt}}^{\mu} \cdot \text{Trt} + \beta_{\text{spp}}^{\mu} \cdot \text{Spp} + \beta_{\text{trt:spp}}^{\mu}(\text{Spp} \times \text{Trt}) + \alpha_{j[i]}$$

$$\alpha_j \sim N\left(0, \sigma_j^2\right)$$

$$\log(\phi_i) = \beta_0^{\phi} + \beta_{\text{spp}}^{\phi} \cdot \text{Species}_i$$

$$\text{logit}(\pi_i) = \beta_0^{\pi} + \beta_{\text{spp}}^{\pi} \cdot \text{Species}_i$$

.

- Y_i: Total basal area per hectare for each species in a 4-meter radius regeneration plot
- π_i: Probability that $Y_i > 0$
- μ_i: Mean of positive values
- ϕ_i: Dispersion parameter for Gamma distribution
- α_j: Group effect pertaining to a site × species interaction

2.4.1.2 Douglas-fir seedling counts

Counts of regenerating Douglas-fir seedlings per vegetation plot were analyzed for differences between harvest treatments using a negative binomial response with a log link, fixed effects for treatment, random effects for sites and macro plots within sites.

$$Y_i \sim \text{NB1}(\mu_i, \phi); \quad \text{where} \quad Var(Y_i) = \mu_i + \phi\mu_i$$

$$\log(\mu_i) = \beta_0 + \beta_{\text{trt}} \cdot \text{Trt}_i + \alpha_{j[i]} + \alpha_{k[i]}$$

$$\alpha_j \sim \text{N}(0, \sigma_j^2), \quad \alpha_k \sim \text{N}(0, \sigma_k^2)$$

.

- Y_i: Count of Douglas-fir seedlings in a 4-meter radius vegetation plot.
- μ_i: The expected count for vegetation plot i.
- β_{trt}: Fixed effect for harvest treatment.
- α_j: Random intercept for site.
- α_k: Random intercept for plots within sites.

2.4.2 Sprout heights

Two metrics were analyzed from the sprout height data: height increment (m year^{-1}), and total height (m) at year 10.

In both models, a random slope for species is included for each macro plot. This specifies a general (unstructured) covariance matrix that allows the random effect variance to differ by species (heteroscedasticity) and allows for non-zero correlations between species-level effects within the same plot (e.g., assessing if plots that are favorable for one species are also favorable for others). This variance-covariance matrix takes the following form, where A, B, and C represent different species:

$$\boldsymbol{\Sigma}_{\text{macro}} = \begin{pmatrix} \sigma_A^2 & \rho_{AB}\sigma_A\sigma_B & \rho_{AC}\sigma_A\sigma_C \\ \rho_{AB}\sigma_A\sigma_B & \sigma_B^2 & \rho_{BC}\sigma_B\sigma_C \\ \rho_{AC}\sigma_A\sigma_C & \rho_{BC}\sigma_B\sigma_C & \sigma_C^2 \end{pmatrix}$$

.

2.4.2.1 Height increment

For height increment, there were two measurements for each tree: increment for years 1 to 5 and increment for years 5 to 10. Thus, analysis of these data included tree as a candidate random effect. While not making a difference for analysis results, it is worth noting that these data were structured differently from the other datasets in that plot was made a globally unique identifier, thus making explicit the nesting of plots within sites or treatments and eliminating the need to specify this nesting structure in the R code model formulas.

The selected height increment model used a normal response distribution on the identity link. It included treatment, growth period, species, the interaction of species and growth period, and the interaction of treatment and species as fixed effects. The treatment/species interaction was included not because the interaction was significant, but because it represented a fundamental research question. A random intercept was included for tree (multiple observations) and macro-plot, and another random effect allowed the response to vary by species differently for each macro plot. The dispersion parameter for the response was modeled (with a log link) as a function of treatment, growth period, species and all three-way interactions.

$$Y_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$$

$$\mu_i = \beta_0^{\mu} + \beta_{\text{trt}}^{\mu} \cdot \text{Trt}_i + \beta_{\text{year}}^{\mu} \cdot \text{Year}_i + \beta_{\text{spp}}^{\mu} \cdot \text{Spp}_i + \beta_{\text{spp:trt}}^{\mu}(\text{Spp}_i \times \text{Trt}_i)$$

$$+ \beta_{\text{spp:year}}^{\mu}(\text{Spp}_i \times \text{Year}_i) + \alpha_{\text{tree}[i]} + \alpha_{\text{macro}[i],\text{spp}[i]}$$

$$\log(\sigma_i) = \beta_0^{\sigma} + \beta_{\text{trt}}^{\sigma} \cdot \text{Trt}_i + \beta_{\text{yr}}^{\sigma} \cdot \text{Year}_i + \beta_{\text{spp}}^{\sigma} \cdot \text{Spp}_i + \cdots + \beta_{\text{3way}}^{\sigma}(\text{Trt}_i \times \text{Year}_i \times \text{Spp}_i)$$

$$\alpha_{\text{tree}} \sim N(0, \sigma_{\text{tree}}^2)$$

$$\alpha_{\text{macro,spp}} = \begin{pmatrix} \alpha_{\text{macro}_1,\text{spp}_1} \\ \alpha_{\text{macro}_1,\text{spp}_2} \\ \vdots \\ \alpha_{\text{macro}_j,\text{spp}_k} \end{pmatrix} \sim \text{N}(\mathbf{0}, \mathbf{\Sigma}_{\text{macro}})$$

.

- Y_i: Height increment (m yr^{-1}) for sprout observation i
- μ_i: Mean height increment for observation i
- σ_i^2: Variance of height increment for observation i
- $\alpha_{\text{tree}}^{\mu}$: Random intercept for tree
- $\alpha_{\text{macro,spp}}^{\mu}$: Random effect for species, co-varying within macro plot
- $\mathbf{\Sigma}_{\text{macro}}$: Unstructured variance-covariance matrix for species within macro plot

2.4.2.2 Height at year 10

Sprout heights at year 10 were modeled with a normal response and a log link. The model with the lowest AIC included species and treatment, but no interactions in the fixed effects. I included the interaction of species and treatment because it was central to our question. I also included a model for dispersion (log link) with predictors species, treatment, and their interaction:

$$Y_i \sim \text{N}(\mu_i, \sigma_i^2)$$

$$\log(\mu_i) = \beta_0^{\mu} + \beta_{\text{trt}}^{\mu} \cdot \text{Trt}_i + \beta_{\text{spp}}^{\mu} \cdot \text{Spp}_i + \beta_{\text{spp:trt}}^{\mu}(\text{Spp} \times \text{Trt}) + \alpha_{\text{spp}[i],\text{macro}[i]}$$

$$\log(\sigma_i) = \beta_0^{\sigma} + \beta_{\text{trt}}^{\sigma} \cdot \text{Trt}_i + \beta_{\text{spp}}^{\sigma} \cdot \text{Spp}_i + \beta_{\text{trt:spp}}^{\sigma}(\text{Trt}_i \times \text{Spp}_i)$$

$$\alpha_{\text{spp,macro}} = \begin{pmatrix} \alpha_{\text{spp}_1,\text{macro}_1} \\ \alpha_{\text{spp}_1,\text{macro}_2} \\ \vdots \\ \alpha_{\text{spp}_j,\text{macro}_k} \end{pmatrix} \sim \text{N}(\mathbf{0}, \mathbf{\Sigma}_{\text{spp,macro}})$$

.

- Y_i: Height at year 10 for tree i
- μ_i: Mean height at year 10 for tree i
- σ_i^2: Variance of height at year 10 for tree i
- $\alpha_{\text{macro,spp}}$: Random effect for species, co-varying within macro plot
- $\mathbf{\Sigma}_{\text{macro,spp}}$: Unstructured variance-covariance matrix for species within macro plot

2.4.3 Fuels

Fuel load in Mg ha^{-1} was estimated for 1-hr, 10-hr, 100-hr, combined duff and litter ("duff/litter"), 1,000-hr, and live vegetation fuels at the transect level. Any transect where a fuel type was not observed was assigned a value of zero for that class and transect.

Fuel load for 1- through 1,000-hr fuels were calculated using the method in Brown (1974). The required size-class specific parameters (specific gravity, particle inclination, and average diameter) were derived from a previous study where I used parameters averaged across all the measured plots (Glebocki, 2015). For 1,000-hr fuels, decay classes 1-3 were considered "sound" and decay classes 4 and 5 were considered "rotten" for the purpose of calculating load and these loads were combined.

Duff and litter were measured at two locations along each transect and these were averaged at the transect level. We did not measure the bulk density of duff and litter fuels but based on a suggestion in another study we used an average combined bulk density (7.73 Mg ha^{-1}) derived from average bulk density across other studies in the redwood region (Finney & Martin, 1993a; Kittredge, 1940; Stuart, 1985). We opted to combine duff and litter components to simplify analysis and focus on total forest floor fuel load.

Vegetation fuel load was calculated using bulk density constants of 8 Mg ha^{-1} m^{-1} for herbaceous fuels and 18 Mg ha^{-1} m^{-1} for woody components (Gangi, 2006). Live and dead vegetation components were combined to calculate herbaceous and woody loading and herbaceous and woody fuel loads were also combined and averaged at the transect level (two measurements per transect) to determine total vegetation loading. Vegetation fuel classes were combined to simplify analysis and because the herbaceous fuel load was generally very low.

Gamma distributed, linear multi-level models, with a log link were used to model fuel load response (Mg ha^{-1}) for all fuel classes. Random intercepts were specified for three levels of nesting, representing sites, macro plots, and transect corners. A hurdle component with logit link was used for predicting zeros in all models except for the duff & litter model. Dispersion components were included where needed, based on AIC model selection. Models were fit separately for pre- and post-PCT fuel measurements. Duff and litter were not measured post-PCT, and vegetation difference was only modeled post-PCT.

$$Y_i \sim \text{hurdle_gamma}(\pi_i, \mu_i, \phi_i)$$

$$\log(\mu_i) = \beta_0^{\mu} + \beta_{\text{trt}}^{\mu} \cdot \text{Trt}_i + \alpha_{\text{site}[i]} + \alpha_{\text{macro}[i]} + \alpha_{\text{corner}[i]}$$

.

Any model component not explicitly defined below was modeled with only an intercept:

$$\text{logit}(\pi_i) = \beta_0^{(\pi,\cdot)}$$

$$\log(\phi_i) = \beta_0^{(\phi,\cdot)}$$

.

- Y_i: Fuel load (Mg ha^{-1}) for transect i
- π_i: Probability that $Y_i > 0$, the hurdle component
- μ_i: Mean of positive values
- ϕ_i: Dispersion parameter for Gamma distribution
- $\alpha_.$: Random intercepts for site, macro plot, and corner
- $\beta_.$: Fixed effects for intercepts and treatment levels for mean, hurdle, and dispersion components

2.4.3.1 Pre-pct

For the Duff & Litter model, there was no hurdle component because there were no zeros in the data, and the dispersion parameter was modeled with only an intercept:

$$\text{logit}\left(\pi_i^{\text{dufflitter}}\right) = \beta_0^{(\pi,\text{dufflitter})}$$

.

For the 10-hr fuel model, the hurdle and the dispersion components were modeled as functions of treatment:

$$\text{logit}\left(\pi_i^{\text{10hr}}\right) = \beta_0^{(\pi,\text{10hr})} + \beta_{\text{trt}}^{(\pi,\text{10hr})} \cdot \text{Trt}_i$$

$$\log\left(\phi_i^{\text{10hr}}\right) = \beta_0^{(\phi,\text{10hr})} + \beta_{\text{trt}}^{(\phi,\text{10hr})} \cdot \text{Trt}_i$$

.

2.4.3.2 Post-pct

After PCT, duff & litter was not modeled, vegetation difference was added.

Duff and litter are not reported for post-pct stands for two reasons. First, these are not expected to have changed significantly given the relatively short time span

before and after PCT. Second, the sampling protocol did not make it clear how to quantify the loading for leaves attached to recently cut branches.

Model selection resulted in somewhat more complex models for dispersion (ϕ) and hurdle (π) components. For 1-hr fuels the dispersion component included treatment and site predictors:

$$\log\left(\phi_i^{1\text{hr}}\right) = \beta_0^{(\phi,1\text{hr})} + \beta_{\text{trt}}^{(\phi,1\text{hr})} \cdot \text{Trt}_i + \beta_{\text{site}}^{(\phi,1\text{hr})} \cdot \text{Site}_i$$

.

For 100-hr fuels, the dispersion and hurdle components included treatment and site predictors:

$$\log\left(\phi_i^{100\text{hr}}\right) = \beta_0^{(\phi,100\text{hr})} + \beta_{\text{trt}}^{(\phi,100\text{hr})} \cdot \text{Trt}_i + \beta_{\text{site}}^{(\phi,100\text{hr})} \cdot \text{Site}_i$$

$$\text{logit}\left(\pi_i^{100\text{hr}}\right) = \beta_0^{(\pi,100\text{hr})} + \beta_{\text{trt}}^{(\pi,100\text{hr})} \cdot \text{Trt}_i + \beta_{\text{site}}^{(\pi,100\text{hr})} \cdot \text{Site}_i$$

.

The dispersion and hurdle components for the rest of the fuel models included only an intercept.

Dispersion models with treatment as the only predictor were included for 1-hr and 100-hr fuel classes. All models included a hurdle portion to model zeros using a logit link. For 100-hr fuels, the hurdle portion included treatment and site as predictors. For the rest, a constant rate of zeros for all observations was used.

3. RESULTS

3.1 Regeneration

3.1.1 Basal area density

According to predictions made from this model, there was not enough evidence to support that treatments resulted in different basal area densities of regeneration. This was true for species individually and when considered together (data not shown). Although trends in means follow the expected pattern of greater regeneration with less retention.

Redwood regeneration showed the strongest apparent treatment response, with the greatest basal area observed in the GS treatment. LD and HD treatments were intermediate, while HD tended to have the lowest redwood basal area.

Tanoak regeneration basal area was intermediate relative to redwood and Douglas-fir. GS and LD treatments showed similar, comparatively higher responses, whereas HA and HD tended to be lower.

Douglas-fir regeneration was low across all treatments, with little apparent variation among them. GS showed the highest basal area, but overall differences were minimal.

Other species (grand fir, madrone, and California wax-myrtle) occurred at low densities across plots. Most plots contained few individuals, though one LD macro plot had notably higher counts. Basal area for other species was low across treatments

overall, with HD tending to be highest and HA lowest, while GS and LD were intermediate (Figure 3).

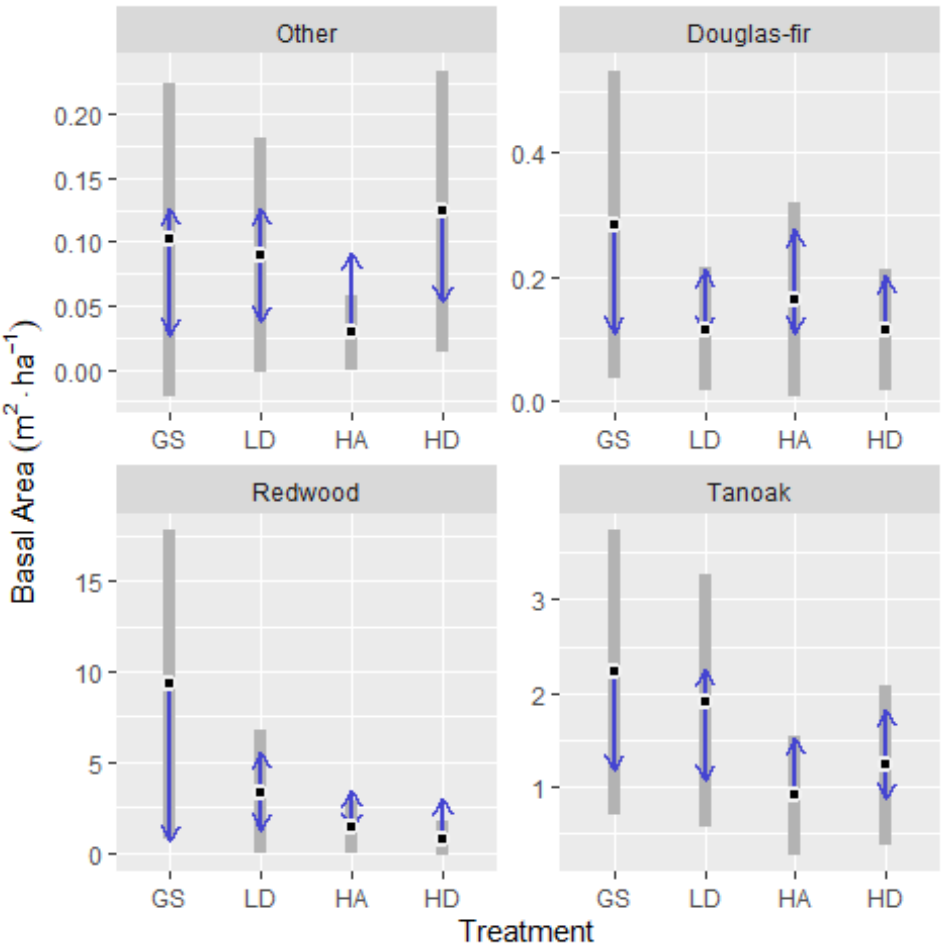


Figure 3. Basal area density (m^2 ha^{-1}) modeled at the vegetation plot level for four harvest treatments and four species classes (n = 16). Gray bars represent the asymtotic 95% confidence interval (α = 0.05), black dots indicate the mean, and blue arrows provide a means of assessing the statistical significance of pairwise differences among treatments within a species (panels). Arrows are drawn so that when two arrows just meet, the p-value for that difference is 0.05 and overlapping arrows indicate p-values greater than 0.05. Note that Y-axes vary by species. While none of the differences were statistically supported, basal area was greatest in the GS treament, varied by species in the LD, and was lowest in the HA and HD. Additionally, redwood basal area was the greatest, tanoak was lower, and Douglas-fir and Other were much lower, but differences were smaller with HA and HD treatments.

Figure 4 highlights the rapid decline in redwood basal area density relative to tanoak with increasing overstory retention. It also illustrates the greater uncertainty

associated with redwood basal area density in the GS treatment, as reflected by the wider confidence intervals compared to those of tanoak.

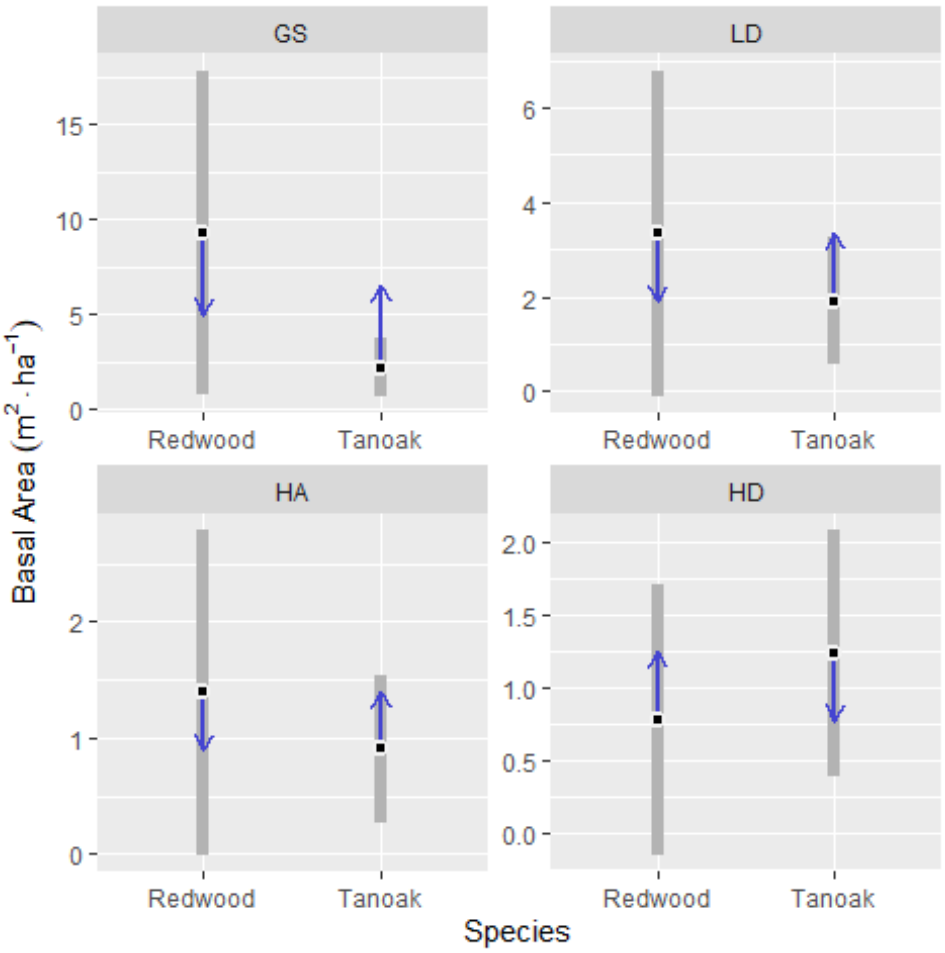


Figure 4. Basal area density (m^2 ha^{-1}) modeled at the vegetation plot level for four harvest treatments and two species classes (n = 16). Gray bars represent the 95% confidence interval (asymptotic, based on normal approximation), black dots—the mean, and non-overlapping blue arrows signify statistical significance (α = 0.05) of comparisons between species redwood and tanoak within a treatment. While none of the differences were statistically supported, this figure highlights the decreasing difference between redwood and tanoak with increasing overstory retention.

3.1.2 Douglas-fir counts

The selected model for Douglas-fir counts did not indicate any statistically significant differences between treatments. Generally, we expect about 2 seedlings per 4-meter-radius plot, or about 400 seedlings per hectare (Figure 5).

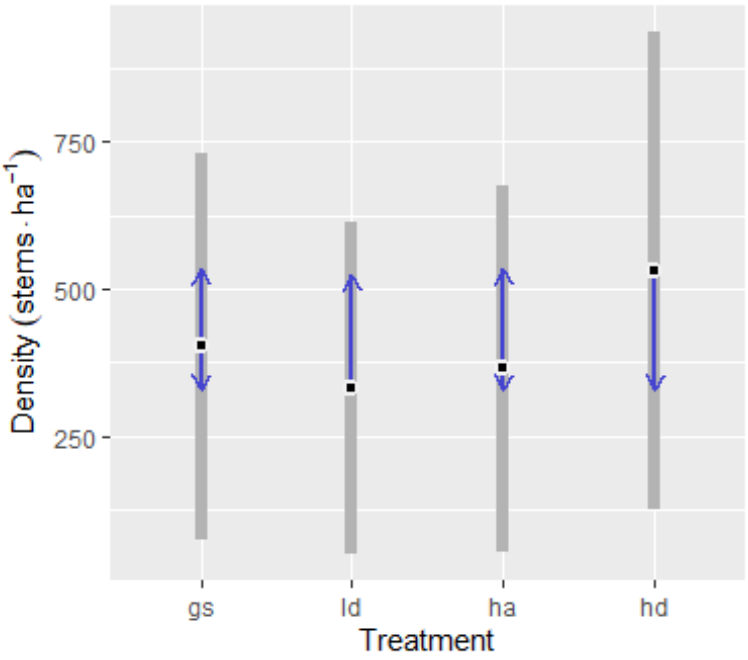


Figure 5. Expected marginal means of vegetation plot level counts of regenerating Douglas-fir seedlings in four harvest treatments 10 years after harvest (n = 16). Results have been scaled to stems per hectare (4-meter radius plots). The asymptotic 95% confidence intervals are based on the normal approximation. Comparisons among treatment were not significant and centered around 400 seedlings per hectare.

3.2 Sprout heights

3.2.1 Height increment

The selected model includes a treatment × species interaction, although this interaction was not statistically significant at the α=0.05 level, it was important for our study questions. I did not include a species × year interaction.

The inclusion of treatment factors in the model ($0.001 \leq p < 0.03$) indicated that the levels of treatment were associated with different growth rates across species and years. And the species x year interaction ($p < 0.001$) suggested that changes in growth rates were different for redwood and tanoak (Figure 6).

For tanoak, height increment was greatest in the GS treatment at 0.47 m yr^{-1}. This was about 0.17 m yr^{-1} more than in the HA and HD treatments, which were very similar at about 0.3 m yr^{-1}.

Redwood followed a similar pattern but with more pronounced differences between treatments. Height increment for redwood in the GS treatment was 0.96 m yr^{-1}, which was about 0.4 m yr^{-1} greater than in the HD treatment ($p = 0$). Additionally, there was evidence that the GS treatment led to greater height increment than the LD treatment by about 0.16 m yr^{-1} ($p = 0.43$). And the LD treatment was higher than the HA treatment by about 0.17 m yr^{-1} ($p = 0.38$).

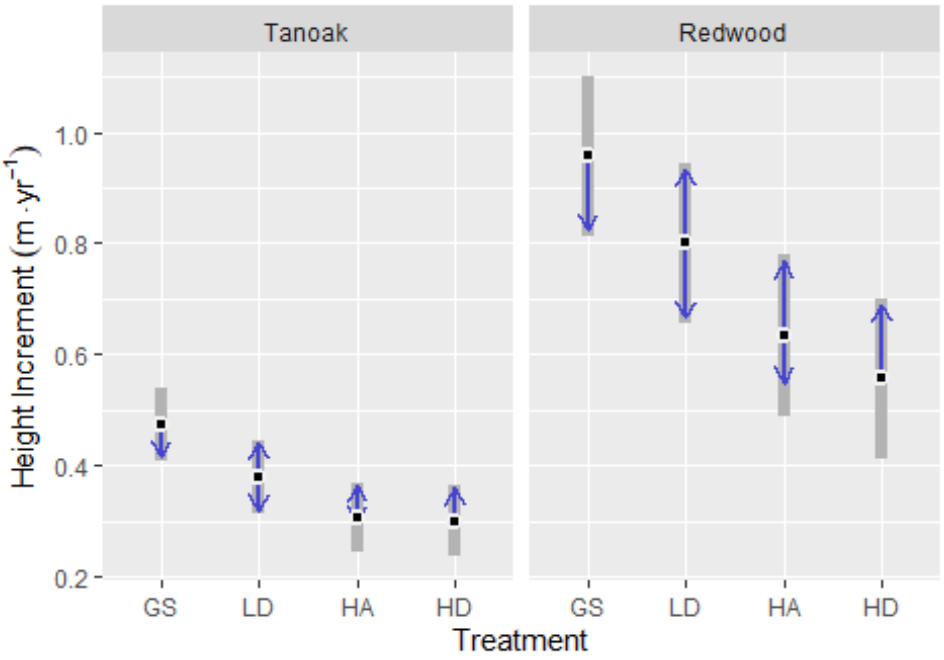


Figure 6. Estimated marginal means for the effect of harvest treatment (x-axis) on redwood and tanoak sprout height increment (y-axis), averaged over two growth periods, ten years after harvest. Gray bars represent asymtotic (normal approximation) confidence intervals and statistical significance (α = 0.05) is indicated by non-overlapping blue arrows for species (panels) between treatments. Height increment decreased with overstory retention were GS > LD > (HA & HD). The difference between GS and HA and HD were statistically supported.

Redwood growth slowed from 0.80 to 0.67 m yr^{-1} in the second period and tanoak slowed from 0.39 to 0.34 m yr^{-1}.

Redwood grew faster than tanoak but slowed down more relative to it in the second period. Height increment for redwood was 0.42 m yr^{-1} greater than tanoak in the first period and 0.33 m yr^{-1} greater than tanoak in the second period (Figure 7).

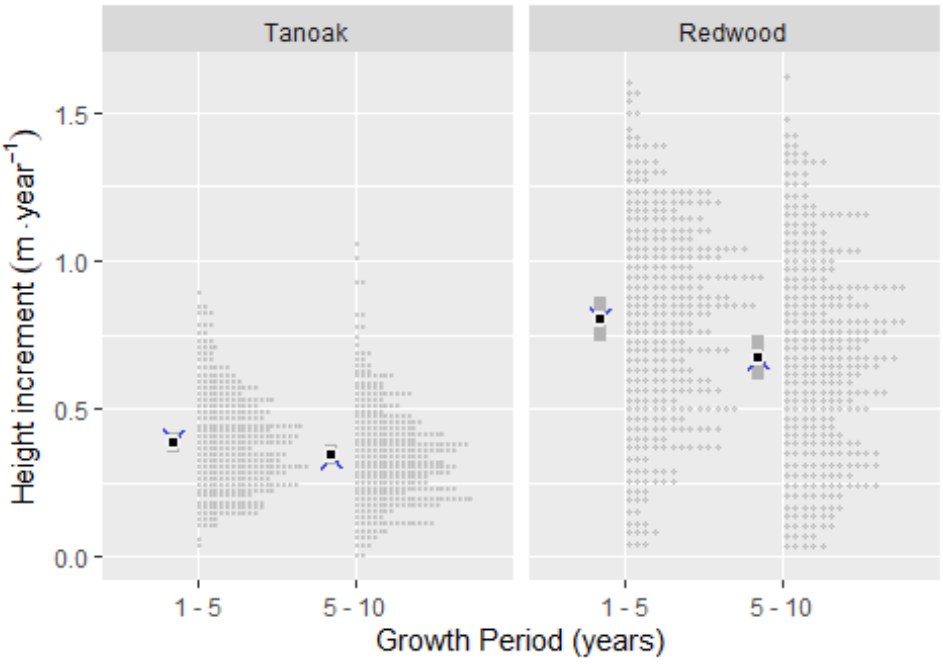


Figure 7. Estimated marginal means for the effect of growth period on redwood and tanoak sprout height increment, averaged over four harvest treatments, from years 1 to 5, and years 5 to 10 after harvest, plotted alongside actual data (small dots). Gray bars represent asymptotic (normal approximation) confidence intervals and statistical significance ($\alpha = 0.05$) is indicated by non-overlapping blue arrows for a period (x-axis) within a growth period (panel). Both tanoak and redwood exhibited a slight decrease in height increment from the first to second periods.

3.2.2 Height at year 10

Because the best model did not contain a species x treatment interaction for the mean response, treatment comparisons were parallel between species. The GS treatment resulted in greater heights in year 10 than the other treatments ($0.001 < p < 0.05$). Predicted mean height for redwood ranged from 10.38 m in the GS treatment to 5.81 m in the HD treatment. For tanoak, predicted mean height ranged from 5.13 m in the GS treatment to 3.12 m in the HD treatment. Predicted mean heights followed the pattern GS > LD > HA ≥ HD (Figure 8).

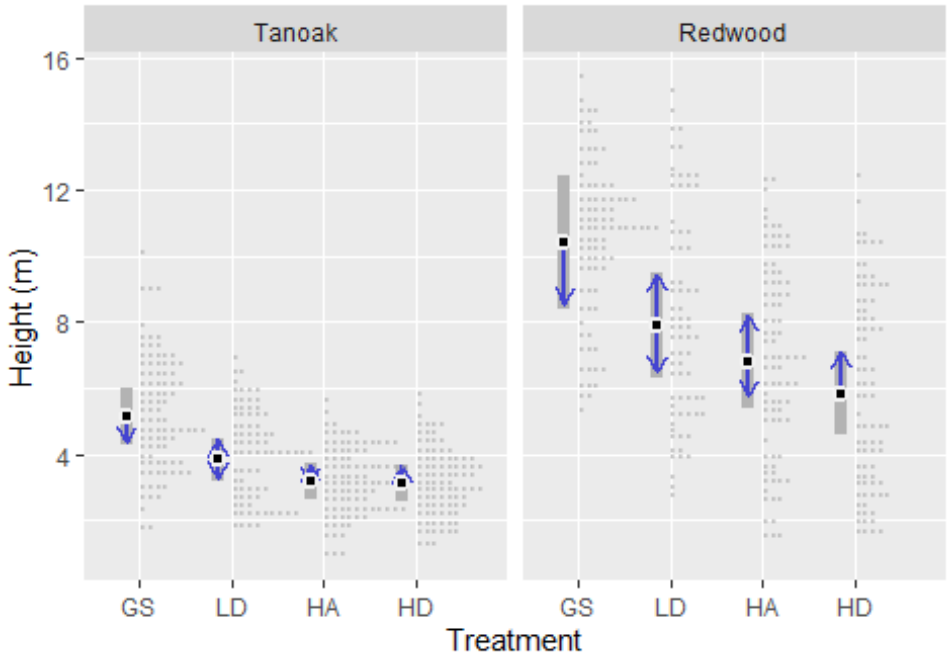


Figure 8. Predicted mean height (black dots) and 95% (asymtotic) confidence intervals (gray bars) for redwood and tanoak stump sprouts 10 years after harvest using four different harvest treatments. Non-overlapping blue arrows indicate statistically significant differences between treatments within a species (panels). Height was greatest in the GS treatment and lowest in the HA and HD treatments. Redwood was taller than tanoak.

3.3 Fuels

3.3.1 Pre-pct

For Duff & Litter, the largest difference was between the HD and HA treatments. The HD treatment had about 54.05 Mg ha^{-1} and was about 14.29 Mg ha^{-1} greater than the HA treatment ($p = 0.09$). Generally, all treatments were similar, with estimated loading of around 47 Mg ha^{-1}.

One-hour fuels were highest in the HA treatment, with an expected value of 1.18 Mg ha^{-1}, which was about Mg ha^{-1} greater than in the GS treatment ($p = 0.03$). One-hour

fuels in the LD and HD treatments were intermediate, but the LD was more similar to the GS and the HD was more similar to the HA treatment.

Ten, hundred and thousand-hour fuels were statistically very similar across treatments ($p \geq 0.7$). Treatment averages had maximum differences of around 1, 3, and 10 Mg ha^{-1} for ten, hundred, and thousand-hour fuels, respectively.

Vegetative fuel loading was greatest in the GS treatment, with an expected value of 28.25 Mg ha^{-1}, which was about 17.88 Mg ha^{-1} greater than in the HA treatment ($p = 0.05$) and LD and HD treatments were intermediate. (Figure 9).

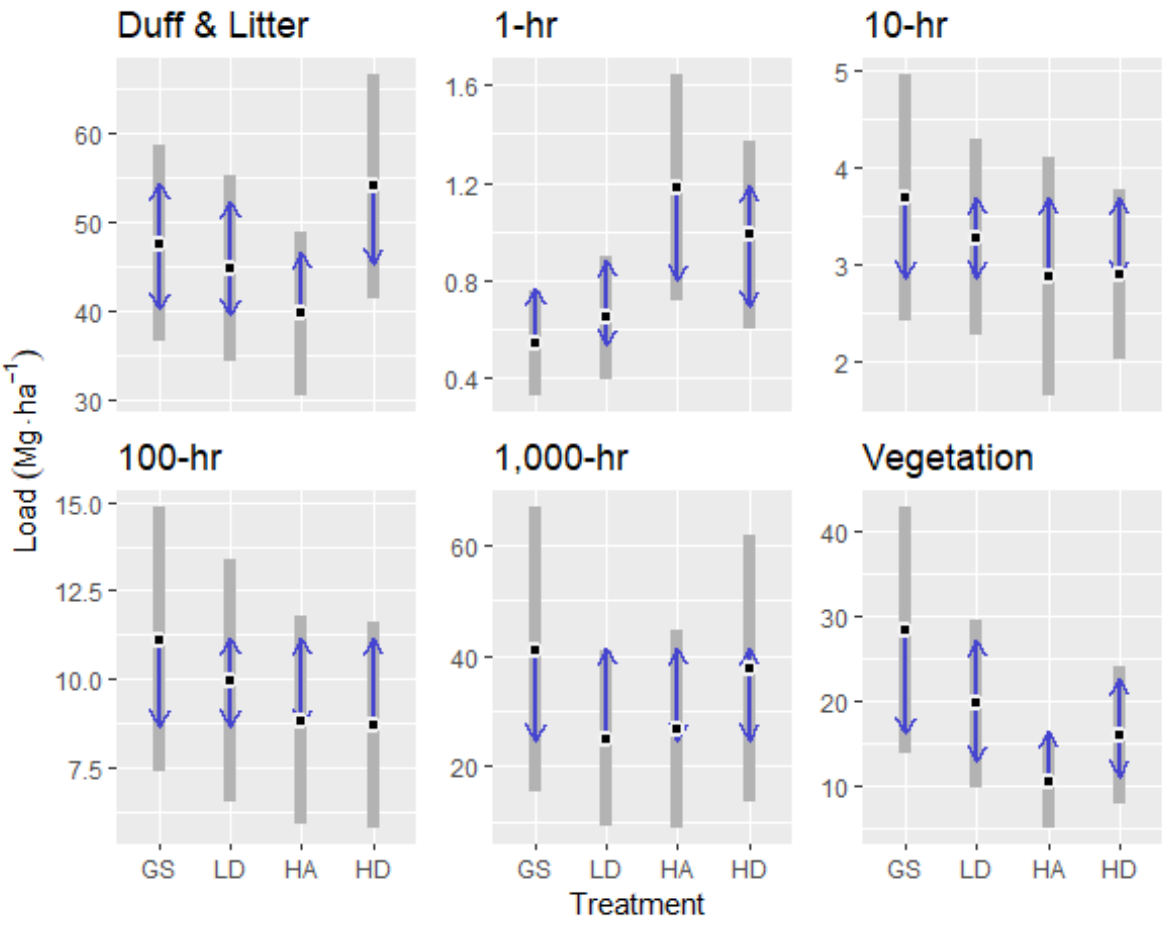


Figure 9. Estimated marginal means (black dots) confidence intervals (gray bands) and comparisons (blue arrows) of surface fuel loading across four treatments for six different fuel-class models. Non-overlapping blue arrows indicates statistical significance at the α = 0.05 level for treatments within a fuel class (panels). Differences between treatments for most classes were small and/or statistically unsupported, except for 1-hr and vegetation fuels for GS and HA treatment comparisons, where 1-hr was lower in the GS treatment and vegetation was higher.

3.3.2 Post-pct

Unlike pre-pct fuel loading which differed little among treatments in terms of fine dead surface fuels, the PCT treatment resulted in additional fine dead surface fuels that differed significantly among treatments. Whereas vegetation fuel loading was reduced and differences between treatments were lessened. Unlike pre-commercial thinning

resulted in greater stratification of treatments (Figure 10). One-hour fuels for most treatments were around 2.5 Mg ha^{-1}, but the HA treatment was lower than these at about 1.34 Mg ha^{-1} ($p \leq 0.01$).

The GS treatment had the greatest 10-hr fuel loading with 8.69 Mg ha^{-1}, which was greater than the LD, HA, and HD treatments by 3.38, 4.97, and 5.73 Mg ha^{-1}, respectively ($p = 0.06$, $p < 0.001$, and $p = 0$). The LD treatment also had about 2.35 Mg ha^{-1} more 10-hr fuels than the HD treatment ($p = 0.03$).

Hundred-hour fuels followed a similar trend to the 10-hr fuels. They were greatest in the GS treatment, with an average of about 17.92 Mg ha^{-1}, which was about 11.03 Mg ha^{-1} greater than the HD treatment ($p = 0.03$).

Thousand-hour fuels were greatest in the HD treatment, with an average of about 59.44 Mg ha^{-1}, which was about 37.06 Mg ha^{-1} greater than the LD and HA treatments ($p \leq 0.14$). The GS treatment was intermediate.

Fuel loading for live vegetation was similar across treatments at around 2.2 Mg ha^{-1}.

The difference in vegetation loading before and after PCT was greatest in the GS treatment at about 29.08 Mg ha^{-1}, which was greater than the HA and HD treatments by about 18 Mg ha^{-1} ($p = 0.09$ and $p = 0.05$, respectively). The LD treatment was intermediate.

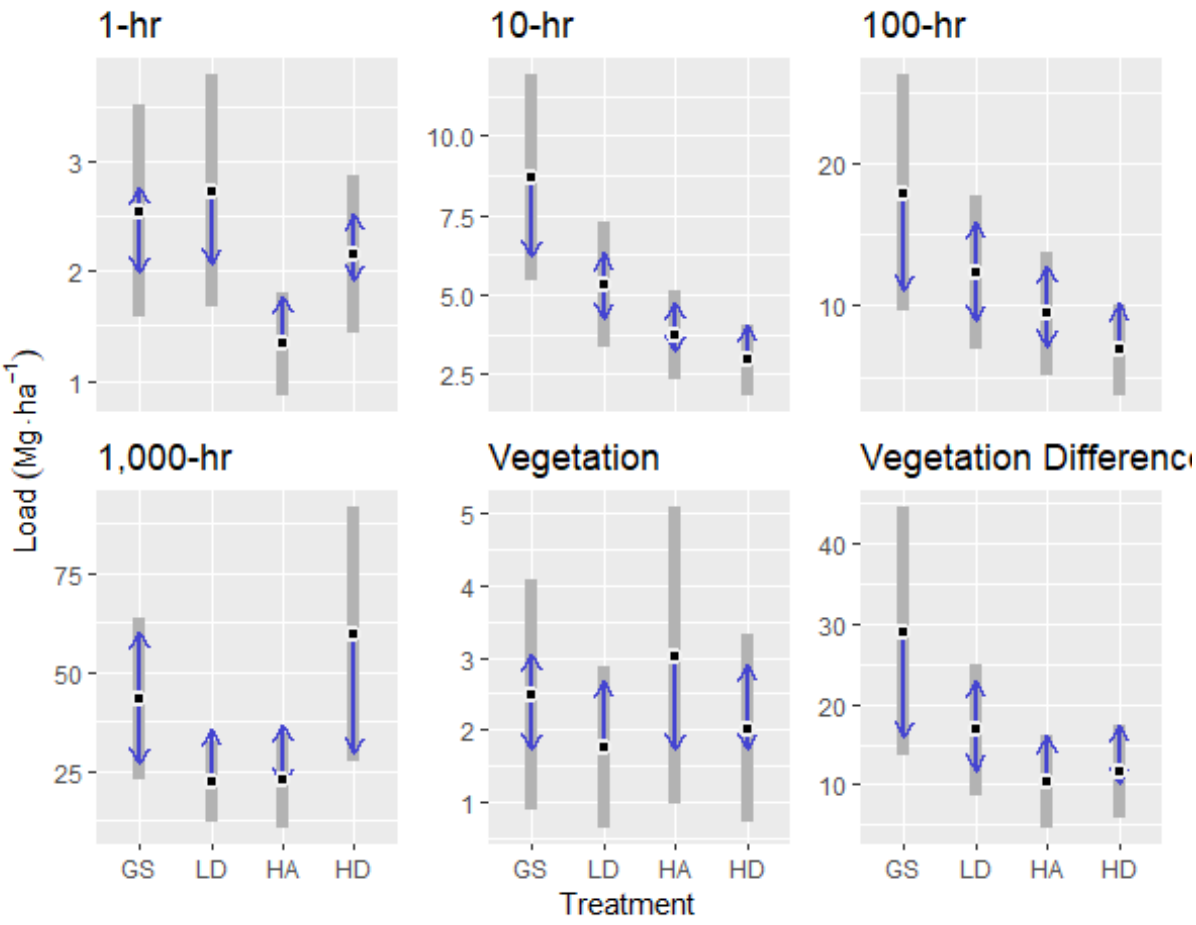


Figure 10. Estimated marginal means (black dots), 95% confidence intervals (asymptotic, gray bars) and statistical comparisons (blue arrows) of fuel loading across four treatments for six different fuel-class models. Non-overlapping blue arrows indicate statistical significance at the α = 0.05 level within a fuel class (panels). Vegetation difference equals the transect level difference in vegetation load in the pre- and post-pct conditions. This represents slash fuels recruited to the forest floor following the pre-commercial thinning. Loading was more differentiated following PCT, where 1-hr, 10-hr, 100-hr, and vegetation difference were all higher in the GS treatment.

3.3.3 Pre-post commercial thinning comparison

The PCT of 10-year-old vegetation dramatically reduced live surface fuels, predictably leading to varying increases in loading across 1- 10- and 100-hr fuels. Specifically, PCT led to a small increase in average 100-hr fuel loading, only for the GS treatment,

increased 10-hr fuels in the GS and LD treatments, and increased 1-hr fuels for all but the HA treatment (Figure 11). However, these results were not statistically comparable, due to slightly different model structures.

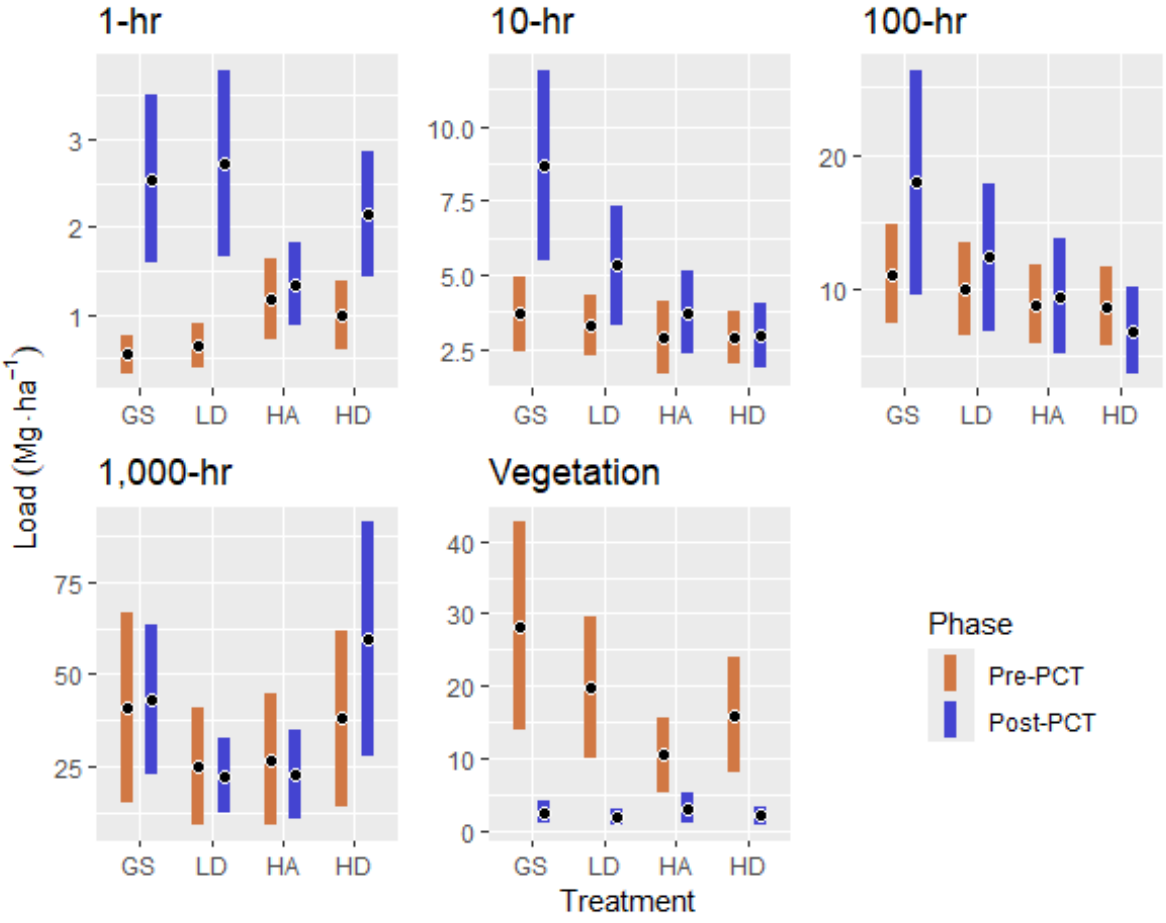


Figure 11. Estimated marginal means (black dots) and confidence intervals (colored bars) of fuel loading across four treatments and five different fuel classes (panels), before and after PCT. Pre- and Post PCT models within a treatment are from similar, but not necessarily identical models. Ninety-five percent (asymptotic) confidence intervals are represented by lighter orange bars for pre-PCT and dark blue for post-PCT fuel loads. This figure shows how fuel loads increase after PCT for 1-, 10-, and to a lesser extent, 100-hr fuels, particularly for the GS treatment, while vegetation loading decreased across all treatments.

4. DISCUSSION

This study corroborates previously documented patterns in redwood–tanoak systems, including the strong positive response of redwood regeneration to canopy openness, the comparatively modest response of tanoak. It validates the notion that 20% relative density is a realistic upper bound for promoting redwood regeneration and provides quantitative evidence of tanoak's shade tolerance. However, given the limited number of treatment replicates, these patterns should be interpreted as strong trends rather than definitive treatment effects.

Similarly, it reiterates the complex dynamics of live and dead surface fuels following harvest and thinning. Fine dead surface fuel loads varied little among treatments prior to pre-commercial thinning. Although minor differences were observed, they were small and unlikely to translate into meaningful differences in fire behavior. In contrast, vegetation fuel loading differed substantially among treatments and contributed to significant differences in fine dead fuels following thinning.

Taken together, these findings highlight the trade-offs inherent in multiaged silviculture. Treatments that promoted vigorous redwood sprout growth also increased live fuel loads and, after thinning, fine dead fuels. Higher levels of overstory retention may approach a threshold at which tanoak's competitive position increases relative to redwood, though uncertainty limits strong inference. Finally, a substantial number of lessons learned, and suggestions will hopefully lead to improved future research.

4.1 Regeneration

4.1.1 Basal area density

While trends in basal area density were generally consistent with those observed for sprout height and expectations regarding canopy openness, the differences among treatments were less pronounced and more variable. This is likely due to a combination of smaller sample sizes, and missing covariates, which are discussed in the limitations section below.

Redwood sprouts basal area density 10 years after harvest treatment responded strongly and positively to increasing levels of openness consistent with previous findings that redwood sprout growth is highly sensitive to light availability and overstory density (Berrill et al., 2018; Muma et al., 2022; O'Hara et al., 2007; O'Hara & Berrill, 2010). In contrast, the tanoak response was comparatively modest reflecting its typical habit of forming a sub-canopy layer in these systems (Hunter et al., 1999; O'Hara et al., 2017).

Shade tolerance theory predicts that at a certain overstory density, understory tanoak abundance should exceed that of redwood. The general trends of these results align with this theory, but the standard errors were too high to derive any clear evidence for this threshold. On the other hand, the general trend of treatment effect on tanaok basal area response was mirrored in the vegetation fuel loading data, which was collected separately, lending support to the idea that higher overstory retention may have minimized tanoak sprout growth.

Average redwood sprouts basal area density 10 years after treatment was much more affected by overstory density than tanoak. Each step increase in overstory density

(GS -> LD -> HA) resulted in average redwood sprout basal area density reductions by around half. Aggregated retention resulted in about 1/3 greater basal area than in the dispersed retention but the p-values for all comparisons were very high ($p >= 0.2$) likely reflecting limited statistical power given the small number of treatment replicates.

Little is understood about tanoak sprout response under shade (Waring & O'Hara, 2008; Wilkinson et al., 1997). This study contributes to that knowledge by describing a range of tanoak sprout development responses with varying overstory density. The results suggest that tanoak sprout basal area density is reduced by about half when moving from open overstory conditions to 20% relative density, however, none of the treatment differences for tanoak were statistically supported.

In all treatments, the uncertainty in the average redwood basal area is much greater than that of tanoak. This suggests that the variability in redwood understory basal area was much greater than that of tanoak. The greater relative uncertainty we observed in redwood compared to tanoak may have to do with redwood's greater sensitivity to light conditions and could indicate a missing covariate to account for this factor. See Limitation and future work below for further discussion of this point.

The sprout basal area density comparisons between redwood and tanoak resulted in steadily decreasing differences between these two species as overstory density increases. This is in contrast to findings from this and other studies when the focus is instead on sprout height growth. The inclusion of basal area here provides a more complete picture of not only sprout vigor but also has implications for total biomass and fuel loading.

Basal area density sampling included relatively few minor species and differences across treatments were small. Though none of the treatment differences were statistically supported, the largest difference occurred between the LD and HA treatment where the latter had less basal area of minor species. While this could be due to low numbers of occurrences of minor species and high variability within treatments, this uniqueness of the HA treatment is found in other places throughout the experiment results: such as for tanoak sprout basal area where the greatest treatment difference was between the GS and HA treatments and in the low vegetation response found in the HA treatment for the pre-PCT fuels data. Potential differences in minor species' abundances are likely only relevant for grand fir, due to its shade tolerance (Webb et al., 2012). Red alder and western hemlock were observed during an earlier iteration of this experiment but were not detected with the current experimental design (Muma, 2019).

4.1.2 Douglas-fir regeneration density

Douglas-fir regeneration abundance was modeled using seedling counts because the species does not regenerate through sprouting and thus did not exhibit the self-thinning found in the sprouting species. Moreover, stems were relatively few in number and consisted largely of small seedlings, making stem density a more informative metric than basal area for comparing responses across treatments.

Douglas-fir is a minor species of special interest due to its commercial value and ability to compete with redwoods in the first 100 years of stand development (Wensel & Krumland, 1986). We did not detect statistically supported differences between treatments in terms of basal area or stem counts. While the observed recruitment density of Douglas-fir seedlings may be sufficient for the eventual production of a viable

cohort in a multiaged system (Schütz and Röhnisch, 2003, as cited in Schütz & Pommerening, 2013), only the LD and GS treatments are likely to maintain overstory densities conducive to Douglas-fir seedling development (Miller & Emmingham, 2001; Schütz & Pommerening, 2013) given the overstory basal areas recorded at the initiation of the experiment (Muma et al., 2022). Rapidly developing redwood and tanoak sprouts will likely further compromise Douglas-fir seedlings competitive ability, especially in the more open treatments where redwood and tanoak sprout growth was more vigorous.

4.2 Sprout height

An important consideration for redwood sprouts in this experiment is that after thinning, a fraction of existing sprouts will be retained. The retained sprouts will generally be the largest and most vigorous. The measurement of total height of each sprout-clump represents the size of the best/tallest sprout, revealing which treatment produced larger individuals that would eventually be selected for retention.

For both redwood and tanoak, height increment and overall height at year 10 were greatest in the GS treatment, followed by LD, HA, and HD treatments. This pattern indicates that greater reductions in overstory density and more understory light lead to more rapid sprout growth.

This strong positive relationship between reduced overstory density and sprout height is a consistent and well-documented finding across various species and forest types in the literature. Understory light availability, which is inversely related to overstory density, is a primary driver of stump sprout growth in redwood-mixed-conifer systems

(Berrill et al., 2018; Berrill et al., 2021), and in other systems as well (Atwood et al., 2009; Gardiner & Helmig, 1997; Keyser & Zarnoch, 2014; Knapp et al., 2017).

Redwood grew faster and achieved greater heights than tanoak across all treatments and growth periods. For example, redwood height increment was 0.42 m/yr greater than tanoak in the first period and 0.33 m/yr greater in the second period. Also, redwood height growth response was much stronger in more open conditions, while tanoak's was comparatively modest.

The increased overstory reduction in this experiment successfully maintained redwood sprout growth, compared to a previous study where overstory reduction was not sufficient to maintain redwood sprout growth (O'Hara & Berrill, 2010). Despite substantial reductions in redwood basal area with the high-overstory-retention treatments, especially compared to tanoak, redwood height growth was still dominant at about 0.6 m yr^{-1}. While the aggregated high-density treatment saw marginally higher sprout height development than the HD treatment, the difference was not large or statistically significant. The initial harvest in both high-density treatments in this experiment resulted in about 39 m^2 ha^{-1}. This suggests that the level of overstory harvest prescribed by a previous growth modeling study (32-38 m^2 ha^{-1}, Berrill & O'Hara, 2009) may be an upper bound in these systems if redwood sprout growth is to be maintained until the next partial harvest.

4.3 Fuel loading

4.3.1 Pre-PCT

Prior to PCT, surface fuel loading was comparable to other studies in redwood systems. The treatment level averages for duff and litter ranged from 40 to 55 Mg ha^{-1} in plots that had post-harvest basal areas (in 2012) of 0 to 40 m^2 ha^{-1}. This was comparable to total duff and litter loading found in 120-year-old redwood stands (range: 29 to 55 Mg ha^{-1}) as well as old-growth stands (average: 50 Mg ha^{-1}) and this study's raw average duff and litter depth of 6.2 cm (data not shown) was similar to that found in a range of redwood stand structures (Finney & Martin, 1993a; Glebocki, 2015; Stuart, 1985).

We used a single average bulk density value from the literature to calculate duff and litter loads, following prior work suggesting that separating the two layers yields only marginal gains in accuracy when estimating total forest floor mass. This approach is particularly justified given the substantial variability reported in duff and litter depths (Finney & Martin, 1993a; Stuart, 1985).

Average total fine woody fuel loading, including one-, ten-, and one hundred-hour-fuels (15 Mg ha^{-1}) was within the range found in 120-year-old stands (9 to 20 Mg ha^{-1}, Finney & Martin (1993b)), double that found in very young, mixed stands (7 Mg ha^{-1}, Glebocki (2015)), and 3 Mg ha^{-1} higher than that found in old growth stands (12 Mg ha^{-1}, Stuart (1985)).

When comparing the average fuel loadings found in this study to those found in the "old growth" and "very young, mixed" stands referenced above, the one-hr fuel loading which, which averaged about 1 Mg ha^{-1} was similar to that found in the old-

growth stand. However, it was about half that found in the very young, mixed stand, possibly due to differing stand structures and species compositions. The very young, mixed stands (Glebocki, 2015) were dominated by young Douglas-fir and would be expected to have different forest floor characteristics and this notion is supported by the finding of similar 1-hr fuel loading as in the old growth study. Another source of possible error is differences in sampling method for 1-hr fuels. We established a cutoff where redwood particles smaller than roughly 2 mm were considered as litter, rather than 1-hr fuels. If we had counted every redwood leaf-spray, regardless of size, we would have likely found higher 1-hr fuel loads.

Average 10-hr fuel loading was higher than in the very young, mixed stand (3.4 vs. 2 Mg ha^{-1}) but similar to the old growth stand. It might be reasonable here to suspect that the smaller trees in the very young, mixed stand did not supply as much 10-hr fuels because of the sizes of their branches, whereas branch shedding patterns in the 10-hr time lags class were more similar between the stands in this study and the old growth ones.

Average 100-hr fuel loading of about 11 Mg ha^{-1} was higher than in both the old growth stands (5 Mg ha^{-1}) and the very young, mixed stands (3 Mg ha^{-1}). This was likely the result of residual fuels (mainly cut tanoak) left over from the harvest treatment that initiated this experiment which would have been absent from the other sites referenced here. In the very young, mixed stand for instance, 100-hr fuels jumped to 10 Mg ha^{-1} immediately following a thinning treatment.

Average 1,000-hr fuel loading of 42 Mg ha^{-1} was somewhat lower than found in old growth stands (63 Mg ha^{-1}) and the very young, mixed stands (54 Mg ha^{-1}). These

estimates are accompanied by relatively high standard errors, but it would not be surprising that State Park forests, with little to no recent harvest activity, would have a larger amount of large downed logs then an actively managed forest such as this.

The similarity in average fuel loading between the Pre-PCT stands and two studies conducted in structurally distinct redwood forests aligns with this study's finding of few statistically significant differences in fuel loading among treatments. This result is common among other studies and is expected due to the highly variable nature of forest fuels (Collins et al., 2016; Keane et al., 2001, 2012).

In a study across California's Sierra Nevada mountain range, fine fuel loading (including 1-, 10-, and 100-hr fuels) were found to be correlated with overstory canopy cover and proportion of shade tolerant species and associated these conditions, which represent recent shifts in forest composition resulting from fire suppression, with an elevated risk of high severity fire (Collins et al., 2016). The differences we observed in duff & litter, and 1-hr fuel loads could be a result of canopy cover and understory species composition directly, but it is likely also a result of the microclimatic differences resulting from these different stand structures. Fine fuel loads are a function of both recruitment and decomposition, and the latter is a function of moisture and temperature (Keane, 2008; Pillers & Stuart, 1993). Forest floor conditions in the HA treatment may have been somewhat moister than the more open treatments due to sheltering but warmer than the HD treatment due to greater light infiltration (e.g., Kane (2021)). While the differences in duff & litter loads we observed are not likely to have a significant impact on fire behavior, small differences in forest floor moisture can have significant implications for prescribed burning operations.

For both 1-hr and vegetation fuel loading, the HA treatment was most similar to the HD treatment. In both treatments it appears that a greater density of overstory trees would result in more twigs and less light available for growth of vegetation at the surface. The high-density treatments had less tanoak BA and it is likely that tanoak was largely responsible for the vegetative fuel loading differences. Vegetation fuel loading was not recorded by species, but redwood and tanoak sprouts composed the vast majority of vegetation fuels.

Understanding of the conditions under which crown fires initiate and spread are not well understood for several reasons and largely because of the difficulty in observing and studying large fires (Finney et al., 2021; Xanthopoulos & Athanasiou, 2020). Less still is known about crown fire behavior in this unique and transitional fuel type (mixed conifer/tanoak sprouts). Because of the relatively low bulk density of canopy fuels, smaller differences in fuel load equate to larger differences in total fuel volume and resulting fire behavior, as well as implications for aerial fuel continuity. These factors warrant more study into the fire behavior of this unique fuel type, including its requirements for combustion, fuel moisture, and more precise characterization of its mass and bulk density.

It was surprising that the modeled vegetation fuel loading differences were not more clearly differentiated because the field crew's experience was that the GS and LD treatments were usually much more difficult to work in and travel through due to greater understory density. An increased fine dead fuel load following PCT was detected, reaffirming the typical trade-off between opening stands to promote growth and the resulting increase in surface fuels (Keyes & Varner, 2006). See below for discussion of

potential shortcomings of the vegetation fuel loading estimation method which may have contributed to this unexpected result.

4.3.2 Post-PCT

Vegetative growth following thinning and harvesting is an important consideration for fire informed management. In this experiment, greater overstory retention resulted in lower understory size and density. Therefore, 10 years of growth (followed by PCT) led to increased average fine woody fuel loading, particularly in the dense understories of the GS and LD treatments. One-hr fuels increased most in the GS and LD treatments but there was also an unexpected increase in 1-hr fuels in the HD treatment compared to the HA treatment which had a negligible increase. Ten-hr fuels increased most in the GS and somewhat in the LD treatment, reflecting the more rapid growth seen in these treatments compared to the others.

Pre-commercial thinning resulted in little change in 1,000-hr fuels, as thinned sapling stems were mostly under three inches, although a potential increase observed in the HD treatment may have been a result of some previously retained canopy trees being cut (12 inches DBH was the prescribed upper limit for thinning). The model selection process resulted in slightly different models after pre-commercial thinning, compared to before thinning for a given fuel class. The model selection process was guided by balancing parsimony, AIC, and the production of well distributed residuals. As predicted fuel load variability increased along with predicted fuel loads following PCT, model selection tended towards more complex models that accounted for this variability–particularly for 1- and 100-hr fuels.

It can be seen from this analysis, resulting surface fuel conditions are expected to be the result of a complex interaction of understory stand productivity, overstory retention, and decomposition rate. Whether any of the differences we observed results in a significant difference in prescribed or wild fire will need to be tested by those disturbances. It is possible that the elevated 10-hr loads in the GS and to a lesser degree, the LD treatments may represent greater fuel continuity which could support prescribed fire operations. Likewise, the HA treatment's lower fuel loading across classes could signify patchy fuel distribution.

4.4 Limitations and future work

A primary limitation of this study relates to the effective sample size for detecting treatment effects. Although a large number of individual sprouts and vegetation subplots were measured, the experimental design included only four replicates per treatment, meaning that treatment-level inference is based on a relatively small number of independent experimental units. As a result, statistical power to detect treatment differences was limited, likely contributing to the large uncertainty and lack of statistical support observed across response variables.

Understory basal area comparisons among treatments and with other species within a treatment for tanoak were confounded by the fact that tanoak produced an abundance of small diameter sprouts, especially in the HA and HD treatments (383 in HA vs 496 in HD, data not shown). Stems smaller than 2.54 cm DBH were tallied. To facilitate basal area analysis, I assigned each of these the possibly inaccurate midpoint

diameter of 1.27 cm. Thus, the basal area of these bushier tanaok in the HD and HA treatments may be slightly artificially elevated.

A key limitation of our modeling approach was the absence of direct measures of understory light availability and thermal environment. Sprout basal area and height, as well as pre-PCT vegetation (which subsequently contributed to post-PCT fine dead fuel loads), are primarily driven by plant productivity and therefore sensitive to variation in light availability. In contrast, surface fuel decomposition is influenced by temperature and microclimatic conditions. Both processes are shaped by canopy structure and topography, which were not explicitly incorporated into the models.

Because the study design included both predominantly north- and south-facing aspects, site-level differences in radiation receipt and heat load were likely bimodal. However, site was modeled as a random effect under the assumption of normality, which may have increased uncertainty in parameter estimates.

Overstory relative density was used as a proxy for understory light but likely did not capture spatial variability in radiation or thermal inputs. Light availability within plots may have been influenced not only by within-plot canopy structure, but also by external shading from upslope trees, road cuts, and broader topographic position. Future work would benefit from integrating direct measures of canopy openness (e.g., hemispherical photography) with modeled insolation or heat-load indices to better characterize total understory energy inputs. Incorporating these measures would likely also improve estimates for important fire-related metrics that were not measured in this study.

Surface fuel moisture and wind speed are two variables which we did not study but have a strong influence on fire behavior and are also affected by overstory density

and arrangement. Given the relatively modest difference we observed in fuel loads, it seems likely that those unmeasured factors may have a greater bearing on fire outcomes and warrant further study in these managed forest systems. Similarly, because duff and litter have different implications for fire behavior, reporting these separately would be an important improvement to future work.

An additional limitation of this study was the method used to estimate vegetation fuel loading. Fuel loads were calculated as estimated percent cover multiplied by estimated height within an assumed cylindrical volume and a constant bulk density. While efficient, this approach may have been imprecise and may have introduced additional uncertainty into estimates of understory biomass. Previous work in redwood systems has successfully estimated understory biomass using regression relationships based on shrub caliper (Stuart, 1985), suggesting that more direct allometric approaches could improve accuracy. Refining estimates of vegetation fuel loading—and incorporating a method to quantify foliage loading in slash fuels generated during PCT—would strengthen the linkage between pre-PCT vegetation biomass and post-PCT fine fuel loading.

Finally, while this study recognizes the importance of incorporating Indigenous perspectives—particularly those of the Northern Pomo, the original stewards of these lands—into the responsible management of keystone species such as tanoak, it did not directly engage with Indigenous perspectives. Future research should prioritize genuine collaboration with Indigenous communities to foster trust, help build capacity, and promote the meaningful co-management of these culturally and ecologically significant ecosystems.

4.5 Conclusions and management implications

This study highlights a fundamental trade-off between multiaged management and fuel hazard reduction. Harvesting treatments that maximized redwood sprout growth also resulted in significantly elevated live fuel loads before pre-commercial thinning. Notably, these responses often represent the same biological phenomenon, with young vegetative growth functioning simultaneously as desired regeneration and as increased surface and ladder fuels, depending on management perspective.

Consequently, following the PCT, the treatments that produced the most vigorous sprouts also generated higher volumes of fine dead fuels. This shows that successful pyrosilviculture (not unlike traditional silviculture) must balance the goals of new cohort development with fuels management. Maximizing the former also increases the volume of slash fuels generated during tending treatments.

This study also confirms that redwood and tanoak respond differently to varying levels of canopy openness. Redwood basal area and height growth responded strongly and positively to increasing openness.

However, the findings suggest that the highest retention treatments approach a critical threshold where tanoak's competitive ability increases relative to redwood. When sprout basal area is used as the measure of competitive advantage, this threshold appeared to occur in the HD treatments, at overstory retentions of 20% relative density, or 39 m^2 ha^{-1}.

For millennia, Indigenous Californians actively shaped redwood–tanoak ecosystems through intentional and frequent use of fire, sustaining food resources,

promoting desired stand conditions, and limiting hazardous fuel accumulation. Tanoak remains culturally central to the Pomo people as a primary food source, emphasizing the need for genuine, collaborative partnerships in future management and research efforts. Integrating Indigenous fire stewardship with contemporary forest management represents not only a pathway toward the practice of pyrosilviculture, but also an essential step toward respectful and sustainable land stewardship.

More broadly, while this study provides an evaluation of how silvicultural treatments influence regeneration dynamics and fuel development, it represents only the initial phase of a much longer experimental effort. Fire behavior and ecological response will ultimately determine whether these treatments achieve their intended outcomes, and each prescribed burn (or wildfire) will constitute a single result shaped by weather, fuel moisture, and ignition conditions. Building a robust understanding of pyrosilviculture will therefore require repeated entries across treatments and time, likely spanning multiple decades.

In contrast, Indigenous stewardship systems, including those of the Northern Pomo, reflect the cumulative outcomes of long-term experimentation with fire over generations. These practices have effectively integrated ecological dynamics that contemporary research is only beginning to quantify. This highlights both the value of long-term experimental studies and the importance of engaging with Indigenous knowledge as complementary sources of understanding.

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APPENDIX — MODELS TABULAR OUTPUTS

A.1 Regeneration

A.1.1 Basal area

Pseduo-code model listing:

Listing A.1

```
Family: Gamma (log)
Conditional: ba_ha ~ treat * spp + (1 | site:spp)
Dispersion: ~spp (log)
Hurdle: ~spp (logit)
```

Table 1. Basal area (m^2 ha^{-1}) modeled at the vegetation plot level for four harvest treatments and four species classes (n = 16). The asymptotic 95% confidence intervals are based on the normal approximation.

Spp	treatment	Emmean	SE	95% LCL	95% UCL
other	gs	0.1	0.06	-0.02	0.22
other	ld	0.09	0.05	-0	0.18
other	ha	0.03	0.01	-0	0.06
other	hd	0.12	0.06	0.01	0.23
df	gs	0.29	0.13	0.04	0.53
df	ld	0.12	0.05	0.02	0.21
df	ha	0.16	0.08	0.01	0.32
df	hd	0.11	0.05	0.01	0.21
rw	gs	9.31	4.36	0.77	17.85
rw	ld	3.34	1.76	-0.11	6.78
rw	ha	1.39	0.72	-0.01	2.8
rw	hd	0.78	0.47	-0.15	1.71
to	gs	2.22	0.78	0.69	3.75
to	ld	1.92	0.69	0.57	3.26
to	ha	0.91	0.32	0.27	1.54
to	hd	1.24	0.43	0.39	2.09

Table 2. Pairwise comparisons of treatments within species. P-values were adjusted using the Tukey method for comparing families of four estimates and they are based on large-sample (asymptotic) normal approximations.

Spp	Contrast	Emmean	SE	P value
other	gs - ld	0.01	0.07	1
other	gs - ha	0.07	0.06	0.6
other	gs - hd	-0.02	0.07	0.99
other	ld - ha	0.06	0.04	0.53
other	ld - hd	-0.03	0.05	0.91
other	ha - hd	-0.1	0.05	0.26
df	gs - ld	0.17	0.12	0.46
df	gs - ha	0.12	0.12	0.76
df	gs - hd	0.17	0.11	0.42
df	ld - ha	-0.05	0.08	0.93
df	ld - hd	0	0.06	1
df	ha - hd	0.05	0.08	0.91
rw	gs - ld	5.97	4.23	0.49
rw	gs - ha	7.92	4.2	0.23
rw	gs - hd	8.53	4.27	0.19
rw	ld - ha	1.95	1.65	0.64
rw	ld - hd	2.56	1.68	0.42
rw	ha - hd	0.62	0.73	0.83
to	gs - ld	0.3	0.69	0.97

to	gs - ha	1.31	0.65	0.18
to	gs - hd	0.98	0.63	0.4
to	ld - ha	1.01	0.57	0.29
to	ld - hd	0.68	0.57	0.63
to	ha - hd	-0.33	0.36	0.8

Table 3. Grand means (m2 ha-1) for basal area of regeneration of each species across treatments 10 years after the initiation of a multiaged redwood forest. The asymptotic 95% confidence intervals are based on the normal approximation.

Spp	Emmean	SE	95% LCL	95% UCL
other	0.09	0.03	0.02	0.15
df	0.17	0.06	0.06	0.28
rw	3.70	1.40	0.96	6.45
to	1.57	0.46	0.66	2.48

A.1.2 Douglas-fir counts

Pseduo-code model listing:

Listing A.2

```
Family: nbinom1 (log)

Conditional: n ~ treat + (1 | site) + (1 | site:treat)
```

Table 4. Vegetation plot level counts of regenerating Douglas-fir seedlings in four harvest treatments 10 years after harvest (n = 16). Results have been scaled to stems per hectare from 4-meter radius plots. The asymptotic 95% confidence intervals are based on the normal approximation.

Treatment	estimate	asymp.LCL	asymp.UCL
gs	402	74	731
ld	331	50	612
ha	366	55	676
hd	531	126	937

A.2 Sprout heights

A.2.1 Height increment

Pseduo-code model listing:

Listing A.3

```
Family: gaussian (identity)

Conditional: ht_inc ~ treat * spp + year * spp + (1 | tree) + (0 + spp | plot)

Dispersion: ~spp * year * treat (log)
```

Table 5. Estimated marginal means for the effect of harvest treatment on redwood and tanoak sprout height increment, averaged over two growth periods, ten years after harvest. The asymptotic 95% confidence intervals are based on the normal approximation.

Spp	Treat	Emmean	SE	95% LCL	95% UCL
TO	GS	0.47	0.033	0.41	0.54
TO	LD	0.38	0.034	0.31	0.45
TO	HA	0.31	0.032	0.24	0.37
TO	HD	0.3	0.032	0.24	0.36
RW	GS	0.96	0.073	0.81	1.1
RW	LD	0.8	0.074	0.65	0.95
RW	HA	0.63	0.074	0.49	0.78
RW	HD	0.56	0.074	0.41	0.7

Table 6. Estimated marginal means for the effect of growth period on redwood and tanoak sprout height increment, averaged over four harvest treatments, from years 1 to 5, and years 5 to 10 after harvest. The asymptotic 95% confidence intervals are based on the normal approximation.

Spp	Year	Emmean	SE	DF	95% LCL	95% UCL
TO	5	0.38	0.017	Inf	0.35	0.42
TO	10	0.34	0.017	Inf	0.31	0.38
RW	5	0.8	0.038	Inf	0.73	0.88
RW	10	0.67	0.038	Inf	0.6	0.75

A.2.2 Height at year 10

Listing A.4

Family: gaussian (log)

Conditional: ht ~ treat * spp + (0 + spp | plot) + (1 | site)

Dispersion: ~spp * treat (log)

Table 7. Height (m) of measured redwood and tanaok sprouts 10 years after harvest treatments with four different over-story densities. The asymptotic 95% confidence intervals are based on the normal approximation.

Spp	Year	Emmean	SE	95% LCL	95% UCL
TO	GS	5.1	0.43	4.3	6
TO	LD	3.8	0.33	3.2	4.5
TO	HA	3.2	0.27	2.6	3.7
TO	HD	3.1	0.27	2.6	3.6
RW	GS	10.4	1.04	8.4	12.4
RW	LD	7.9	0.82	6.3	9.5
RW	HA	6.8	0.73	5.4	8.2
RW	HD	5.8	0.64	4.6	7.1

Table 8. Pairwise comparisons of treatments within species for height (m) of measured redwood and tanoak sprouts 10 years after harvest. The P-values are based on the normal approximation.

Spp	Contrast	Emmean	SE	P value
TO	GS - LD	1.31	0.54	0.08
TO	GS - HA	1.97	0.51	0
TO	GS - HD	2.01	0.51	0
TO	LD - HA	0.66	0.43	0.41
TO	LD - HD	0.7	0.43	0.35
TO	HA - HD	0.04	0.38	1
RW	GS - LD	2.49	1.32	0.23
RW	GS - HA	3.57	1.27	0.02
RW	GS - HD	4.57	1.22	0
RW	LD - HA	1.08	1.1	0.76
RW	LD - HD	2.08	1.04	0.19
RW	HA - HD	1.01	0.97	0.73

A.3 Fuels

Pseduo-code model listing:

Listing A.5

```
family: Gamma(link = "log")

conditional: fuel_load ~ treatment + (1 | site/macro/corner)
```

A.3.1 Pre-pct

Table 9. Dispersion and hurdle component specifications for six fuel classes before pct.

class	Dispersion (log)	Hurdle (logit)
Duff & Litter	~1	~0
1-hr	~1	~1
10-hr	~trt	~trt
100-hr	~1	~1
1,000-hr	~1	~1
Vegetation	~1	~1

Table 10. Estimated marginal means (Mg ha-1) for six fuel classes and four overstory treatments 10 years after partial harvest and prior to pre-commercial thinning (PCT). The asymptotic 95% confidence intervals are based on the normal approximation.

Class	Treat	Emmean	SE	95% LCL	95% UCL
Duff & Litter	gs	47.62	5.66	36.53	58.71
Duff & Litter	ld	44.8	5.32	34.38	55.23
Duff & Litter	ha	39.76	4.72	30.51	49
Duff & Litter	hd	54.05	6.42	41.47	66.63
1-hr	gs	0.54	0.11	0.33	0.76
1-hr	ld	0.64	0.13	0.39	0.9
1-hr	ha	1.18	0.24	0.71	1.64
1-hr	hd	0.98	0.2	0.6	1.37
10-hr	gs	3.68	0.65	2.41	4.94
10-hr	ld	3.28	0.51	2.28	4.29
10-hr	ha	2.88	0.62	1.66	4.1
10-hr	hd	2.89	0.45	2.02	3.77
100-hr	gs	11.1	1.91	7.36	14.84
100-hr	ld	9.97	1.75	6.54	13.39
100-hr	ha	8.84	1.51	5.88	11.8
100-hr	hd	8.71	1.48	5.8	11.62
1,000-hr	gs	41.01	13.18	15.18	66.84
1,000-hr	ld	24.99	8.17	8.98	40.99
1,000-hr	ha	26.71	9.12	8.84	44.58

1,000-hr	hd	37.84	12.32	13.69	61.98
Vegetation	gs	28.25	7.34	13.86	42.64
Vegetation	ld	19.7	5.02	9.86	29.54
Vegetation	ha	10.37	2.7	5.08	15.66
Vegetation	hd	15.91	4.07	7.93	23.89

Table 11. Pairwise comparisons of treatments for six fuel classes before pct. P-values were adjusted for multiple comparisons using the Tukey method and are based on normal approximations.

Class	Contrast	Emmean	SE	P value
1-hr	gs - ha	-0.6343	0.23	0.033
Vegetation	gs - ha	17.8789	7.04	0.054
Duff & Litter	ha - hd	-14.2941	6.12	0.09
1-hr	ld - ha	-0.5315	0.24	0.115
1-hr	gs - hd	-0.4426	0.2	0.119
Vegetation	ld - ha	9.3275	5	0.242
Vegetation	gs - hd	12.34	7.35	0.334
1-hr	ld - hd	-0.3398	0.21	0.353
Duff & Litter	ld - hd	-9.2477	6.34	0.463
Duff & Litter	gs - ha	7.8609	5.6	0.497
Vegetation	ha - hd	-5.5389	4.22	0.555
100-hr	gs - hd	2.3906	2.12	0.671
Vegetation	gs - ld	8.5514	7.66	0.679
1,000-hr	gs - ld	16.0218	15.01	0.71
100-hr	gs - ha	2.2594	2.14	0.715
10-hr	gs - hd	0.7859	0.78	0.748
Duff & Litter	gs - hd	-6.4332	6.49	0.755
Duff & Litter	ld - ha	5.0463	5.39	0.786
1,000-hr	gs - ha	14.2963	15.56	0.795

1,000-hr	ld - hd	-12.8493	14.19	0.802
10-hr	gs - ha	0.7952	0.89	0.809
1,000-hr	ha - hd	-11.1238	14.66	0.873
1-hr	ha - hd	0.1917	0.27	0.889
1-hr	gs - ld	-0.1027	0.15	0.896
Vegetation	ld - hd	3.7886	5.52	0.902
100-hr	ld - hd	1.2551	2.01	0.924
10-hr	ld - hd	0.3889	0.68	0.94
100-hr	ld - ha	1.1238	2.02	0.945
100-hr	gs - ld	1.1355	2.27	0.959
10-hr	ld - ha	0.3982	0.8	0.96
10-hr	gs - ld	0.397	0.82	0.963
Duff & Litter	gs - ld	2.8145	5.89	0.964
1,000-hr	gs - hd	3.1725	17.48	0.998
1,000-hr	ld - ha	-1.7255	11.62	0.999
100-hr	ha - hd	0.1313	1.84	1
10-hr	ha - hd	-0.0093	0.77	1

Table 12. Modeled grand means Mg ha-1 of fuel loading 10 years after harvest and prior to PCT. The asymptotic 95% confidence intervals are based on the normal approximation.

Class	Emmean	SE	95% LCL	95% UCL
Duff & Litter	46.56	4.18	38.35	54.8
1-hr	0.84	0.12	0.61	1.1
10-hr	3.19	0.28	2.63	3.7
100-hr	9.66	1.09	7.52	11.8
1,000-hr	32.63	6	20.87	44.4
Vegetation	18.56	3.32	12.05	25.1

A.3.2 Post-pct

Table 13. Model specifications for six fuel classes after pct.

class	Dispersion (log)	Hurdle (logit)
1-hr	~trt + site	~1
10-hr	~1	~1
100-hr	~trt + site	~trt + site
1,000-hr	~1	~1
Vegetation	~1	~1
Vegetation Difference	~1	~1

Table 14. Estimated marginal means (Mg ha-1) for six fuel classes and four overstory treatments 10 years after partial harvest and after pre-commercial thinning (PCT). The asymptotic 95% confidence intervals are based on the normal approximation.

Class	Treat	Emmean	SE	95% LCL	95% UCL
1-hr	gs	2.5	0.49	1.59	3.5
1-hr	ld	2.7	0.54	1.67	3.8
1-hr	ha	1.3	0.24	0.87	1.8
1-hr	hd	2.1	0.37	1.43	2.9
10-hr	gs	8.7	1.64	5.48	11.9
10-hr	ld	5.3	1.01	3.32	7.3
10-hr	ha	3.7	0.71	2.33	5.1
10-hr	hd	3	0.57	1.85	4.1
100-hr	gs	17.9	4.26	9.57	26.3
100-hr	ld	12.3	2.79	6.87	17.8
100-hr	ha	9.4	2.2	5.12	13.8
100-hr	hd	6.9	1.62	3.7	10.1
1,000-hr	gs	43.1	10.43	22.68	63.6
1,000-hr	ld	22.4	5.24	12.1	32.6
1,000-hr	ha	22.8	6.1	10.85	34.8
1,000-hr	hd	59.4	16.26	27.56	91.3
Vegetation	gs	2.5	0.81	0.9	4.1
Vegetation	ld	1.8	0.57	0.66	2.9
Vegetation	ha	3	1.04	0.98	5.1

Vegetation	hd	2	0.66	0.72	3.3
Vegetation Difference	gs	29.1	7.85	13.71	44.5
Vegetation Difference	ld	16.9	4.15	8.73	25
Vegetation Difference	ha	10.4	2.93	4.65	16.1
Vegetation Difference	hd	11.6	2.96	5.8	17.4

Table 15. Pairwise comparisons of treatments for six fuel classes after pct. P-values were adjusted for multiple comparisons using the Tukey method and are based on normal approximations.

Class	Contrast	Emmean	SE	P value
10-hr	gs - hd	5.73	1.4	0.00026
10-hr	gs - ha	4.97	1.38	0.00172
1-hr	ld - ha	1.38	0.42	0.00585
1-hr	gs - ha	1.21	0.38	0.00889
1-hr	ha - hd	-0.81	0.26	0.00919
100-hr	gs - hd	11.03	3.93	0.02579
10-hr	ld - hd	2.35	0.84	0.02637
Vegetation Difference	gs - ha	18.69	7.35	0.05357
10-hr	gs - ld	3.38	1.36	0.06286
Vegetation Difference	gs - hd	17.47	7.42	0.08602
1,000-hr	ld - hd	-37.06	16.86	0.12359
1,000-hr	ha - hd	-36.63	17.13	0.14104
100-hr	gs - ha	8.48	3.98	0.14298
100-hr	ld - hd	5.46	2.6	0.15262
10-hr	ld - ha	1.59	0.85	0.24322
1,000-hr	gs - ld	20.75	11.4	0.26381
1,000-hr	gs - ha	20.32	11.8	0.31242
Vegetation Difference	gs - ld	12.22	7.54	0.36644
Vegetation Difference	ld - ha	6.47	4.23	0.41898

1-hr	ld - hd	0.58	0.41	0.49237
Vegetation	ld - ha	-1.26	0.92	0.51617
100-hr	gs - ld	5.57	4.07	0.51975
Vegetation Difference	ld - hd	5.25	4.22	0.59869
10-hr	ha - hd	0.76	0.61	0.60281
100-hr	ha - hd	2.55	2.15	0.63653
Vegetation	ha - hd	1	0.91	0.68792
1-hr	gs - hd	0.4	0.36	0.6903
100-hr	ld - ha	2.91	2.74	0.71314
Vegetation	gs - ld	0.72	0.71	0.7434
1,000-hr	gs - hd	-16.31	18.97	0.82556
Vegetation	gs - hd	0.46	0.75	0.92729
Vegetation	gs - ha	-0.54	0.97	0.94512
Vegetation	ld - hd	-0.26	0.62	0.97521
1-hr	gs - ld	-0.18	0.47	0.98118
Vegetation Difference	ha - hd	-1.22	3.46	0.98495
1,000-hr	ld - ha	-0.44	7.65	0.99993

Table 16. Modeled grand means Mg ha-1 of fuel loading 10 years after harvest and PCT. The asymptotic 95% confidence intervals are based on the normal approximation.

Class	Emmean	SE	95% LCL	95% UCL
1-hr	2.2	0.35	1.5	2.9
10-hr	5.2	0.81	3.6	6.8
100-hr	11.6	2.05	7.6	15.7
1,000-hr	36.9	5.52	26.1	47.8
Vegetation	2.3	0.61	1.1	3.5
Vegetation Difference	17	3.27	10.6	23.4